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Credits: This presentation includes slides and images from various books, research articles, and online sources, including:

- NVIDIA's Edge AI and Robotics Teaching Kit
- Harvard - TinyML Courseware and Machine Learning Systems
- MIT - TinyML and Efficient Deep Learning Computing
- Dive into Deep Learning and Luis Serrano's Academy and his books and videos
- Hands-on Machine Learning with Scikit-Learn, Keras and TensorFlow book
- IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things, 2017 by David Hanes
- Introduction to IoT, 2021 by Anandarup Mukherjee; *AI at the Edge* by Daniel Situnayake, Jenny Plunkett - O'Reilly

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Instruction Enhancement Program (IEP) on

Edge AI

Lecture 1: Introduction to IoT, Edge Analytics, and Edge AI

Pandarasamy Arjunan (Samy)
Assistant Professor
Robert Bosch Centre for Cyber Physical Systems
Indian Institute of Science, Bengaluru
samy@iisc.ac.in

Image source: https://leverx.com

Career (Academic and Industry):

- Assistant Professor, IISc, Bangalore (Since July 2023)
- Postdoc Scholar, BEARS, UC Berkely's Lab in Singapore (2018-2023)
- Data Scientist (DataGen and SenSing) (2017-2018)
- IBM Research (Fellowship and Internship), Bangalore (2014-2015)
- Visiting Graduate Researcher, NESL @ UCLA (2013)
- PhD Scholar, IIIT-Delhi

Research Interests and Expertise:

- IoT, CPS, AI/ML and Edge AI
- Domains: Smart X (Energy, Buildings, Cities, Mobility, and Agri)
- Edge Intelligence Lab @ IISc

More details at https://www.samy101.com/

IIITD

IBM Research

BEARS | Berkeley Education Alliance for Research in Singapore Limited

UCLA

Berkeley UNIVERSITY OF CALIFORNIA

SinBerBEST²

Building Efficiency and Sustainability in the Tropics

Lecture Plan

Today

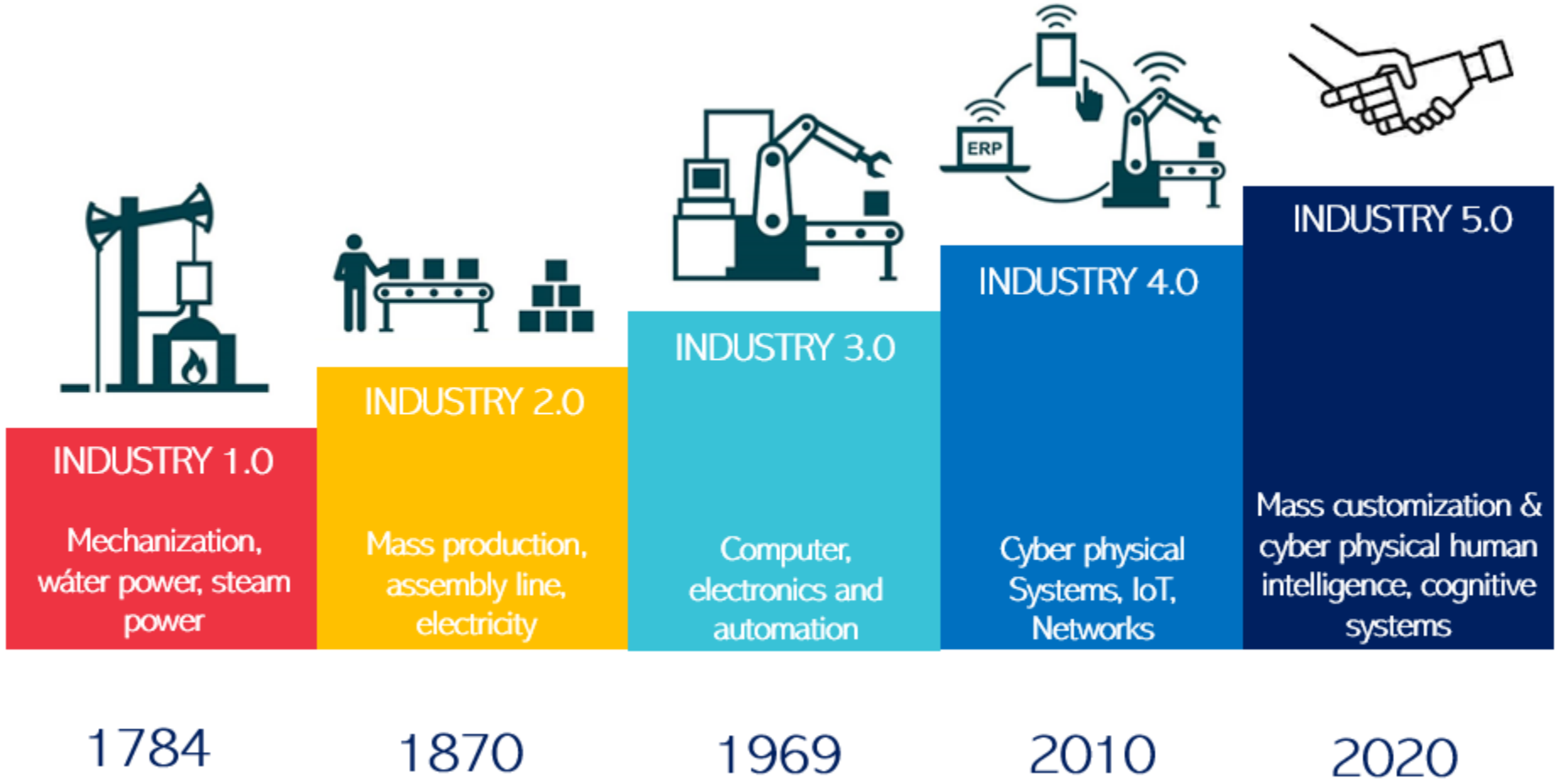
- **Lecture 1**: Introduction to IoT, Edge Analytics, and Edge AI
- **Lecture 2**: Tiny Machine Learning (TinyML)
- **Lab 1**: Human Activity Recognition using TinyML

Tomorrow

- **Lecture 3**: Overview of Embedded Computer Vision
- **Lecture 4**: Efficient Deep Learning and Efficiency Metrics
- **Lecture 5**: Model Optimization Techniques (Quantization, Pruning, etc.)
- **Lab 2**: Embedded Computer Vision
- **Lab 3**: Model Optimization for Edge AI (online)

Lecture 1:
Introduction to IoT, Edge Analytics, and Edge AI

Industrial Revolution



*Years are estimates

Source: Internet

What is Internet of Things (IoT)?

IoT is a network of physical objects that are embedded with sensors, software, and network connectivity, allowing them to collect, exchange, and act on data over the internet.

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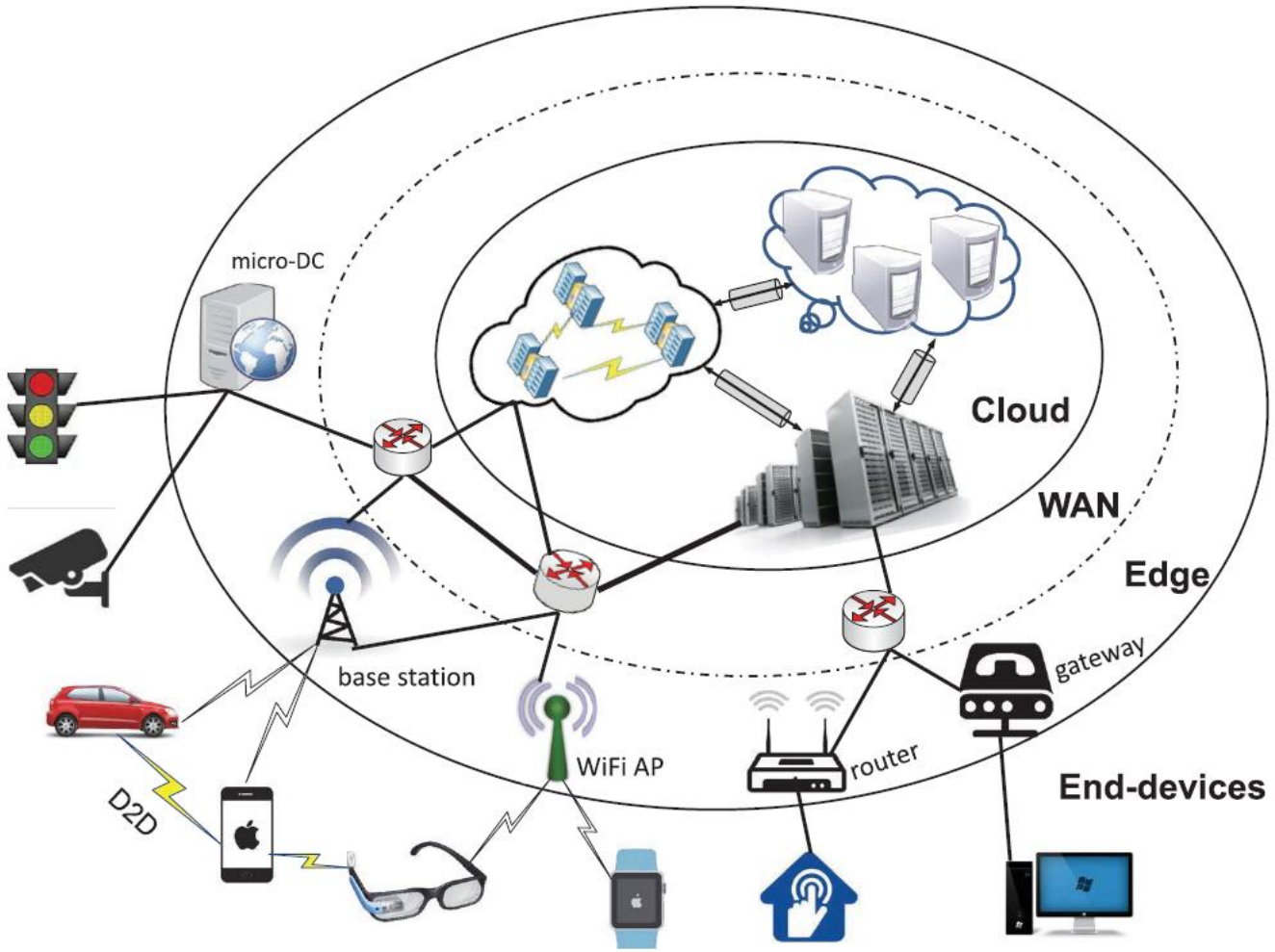
IoT and Edge Devices



Edge Devices

- Provide connectivity between end devices and cloud services
- Deployed close to data sources to enable low-latency processing
- Operate under limited compute, memory, and energy budgets

Edge Computing

- Processes data close to where it is generated (at the network edge), rather than relying solely on centralized cloud data centers
- Reduces latency and bandwidth usage
- Enables real-time processing and decision-making
- Improves privacy, reliability, and energy efficiency

Source: Edge Intelligence: Paving the Last Mile of Artificial Intelligence With Edge Computing (Zhou et al. 2019)

Types of IoT

Consumer IoT: Personal devices like smart homes, wearables, and connected appliances.

Industrial IoT (IIoT): Applications in industries for automation and efficiency, like smart factories.

Commercial IoT: Solutions for businesses, such as smart retail and connected healthcare.

Infrastructure IoT: Large-scale systems for smart cities, traffic management, and energy grids.

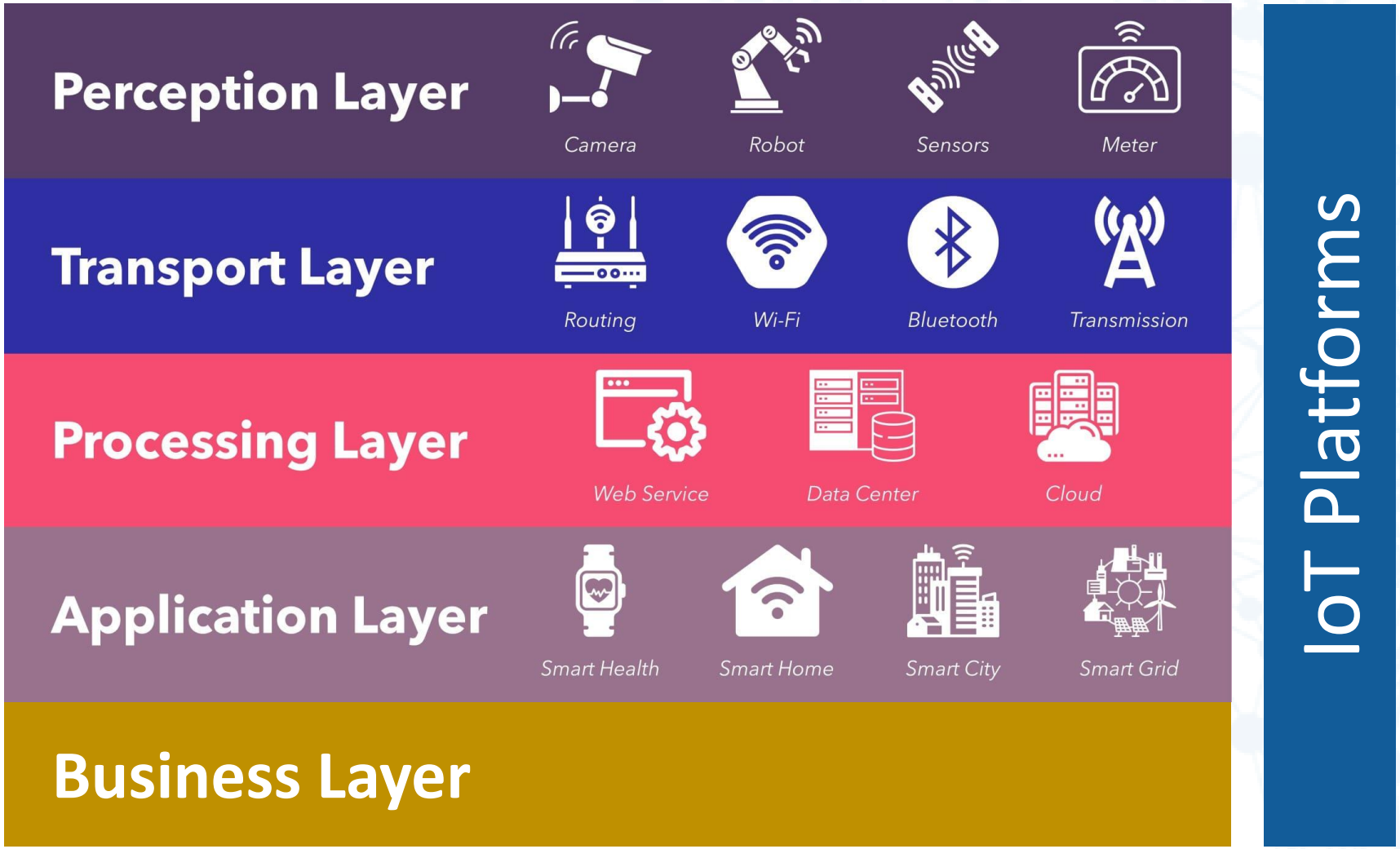
Types of IoT based on different Things or Devices

- Internet of Vehicles (IoV)
- Internet of Drones (IoD)
- Internet of Wearables (IoW)
- Internet of Appliances (IoA)
- Internet of Cameras (IoC)
- Internet of Sensors (IoS)
- Internet of Robots (IoR)
- Internet of Meters (IoM)
- Internet of Wearable Health Devices (IoWHD)
- Internet of Smart Vehicles (IoSV)
- More…

Related Technologies

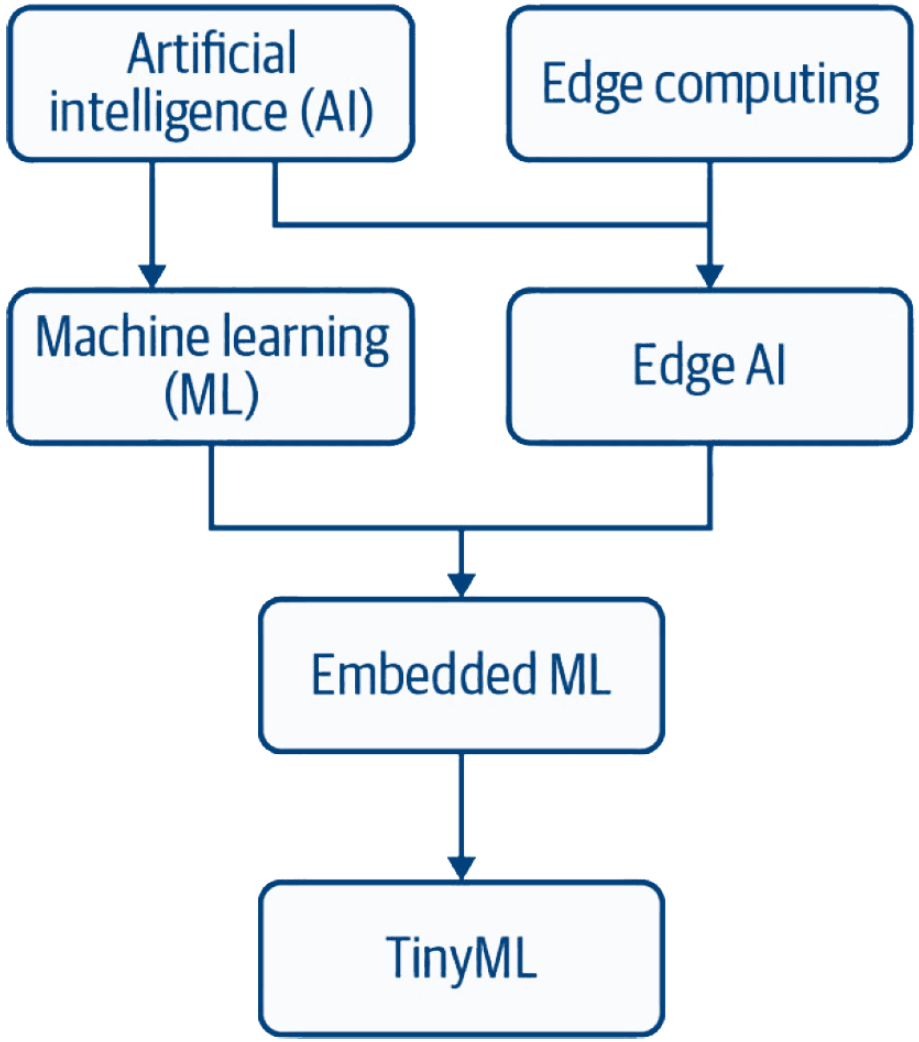
- Web of Things (WoT)
- Machine-to-Machine (M2M)
- Cyber-Physical Systems (CPS)
- Smart Objects
- Industrial Internet (IIoT)
- Ambient Intelligence (AmI)
- Wireless Sensor Networks (WSN)
- Networked Embedded Systems
- Digital Twin
- Fog Computing
- Edge Computing
- 5G/6G Networks

IoT Architecture and Layers



Source: https://dgtlinfra.com/

The BIG picture of Edge AI



Processing data near the data source instead of sending it to the cloud.

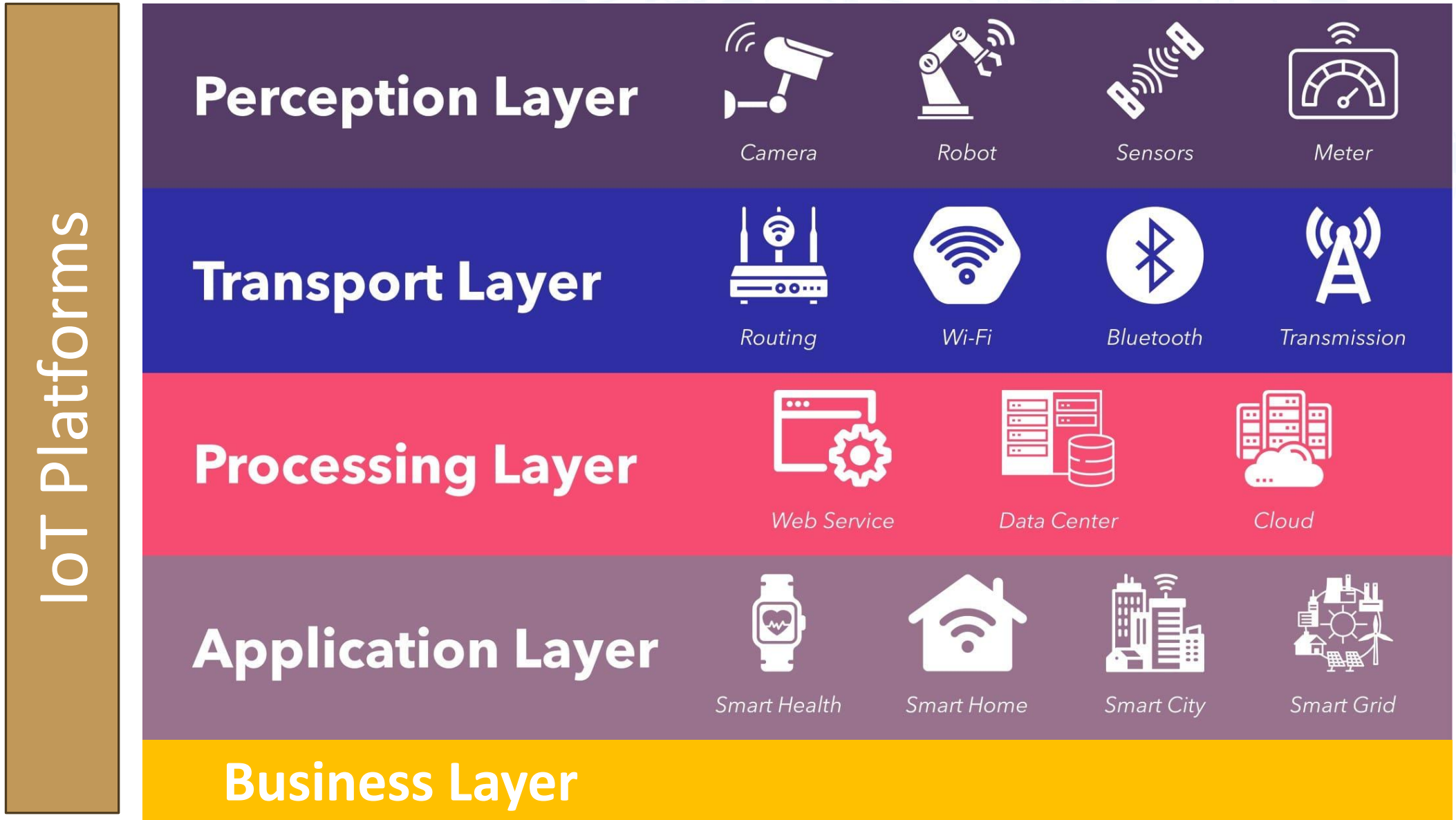
Executing and adapting AI models on edge devices, enabling local intelligence without reliance on the cloud.

Running ML models on resource-constrained embedded systems

Ultra-low-power ML on microcontrollers with extremely limited resources

Source: [3]

IoT Architecture and Layers



Source: https://dgtlinfra.com/

Perception Layer

- It is the physical layer in IoT responsible for sensing, collecting, and transmitting data from the environment.
- Sensors (e.g., temperature, humidity, motion) and Actuators for physical interaction.

Functions:

- Detects physical parameters and converts them into digital signals.
- Gathers real-time data from the environment or devices.

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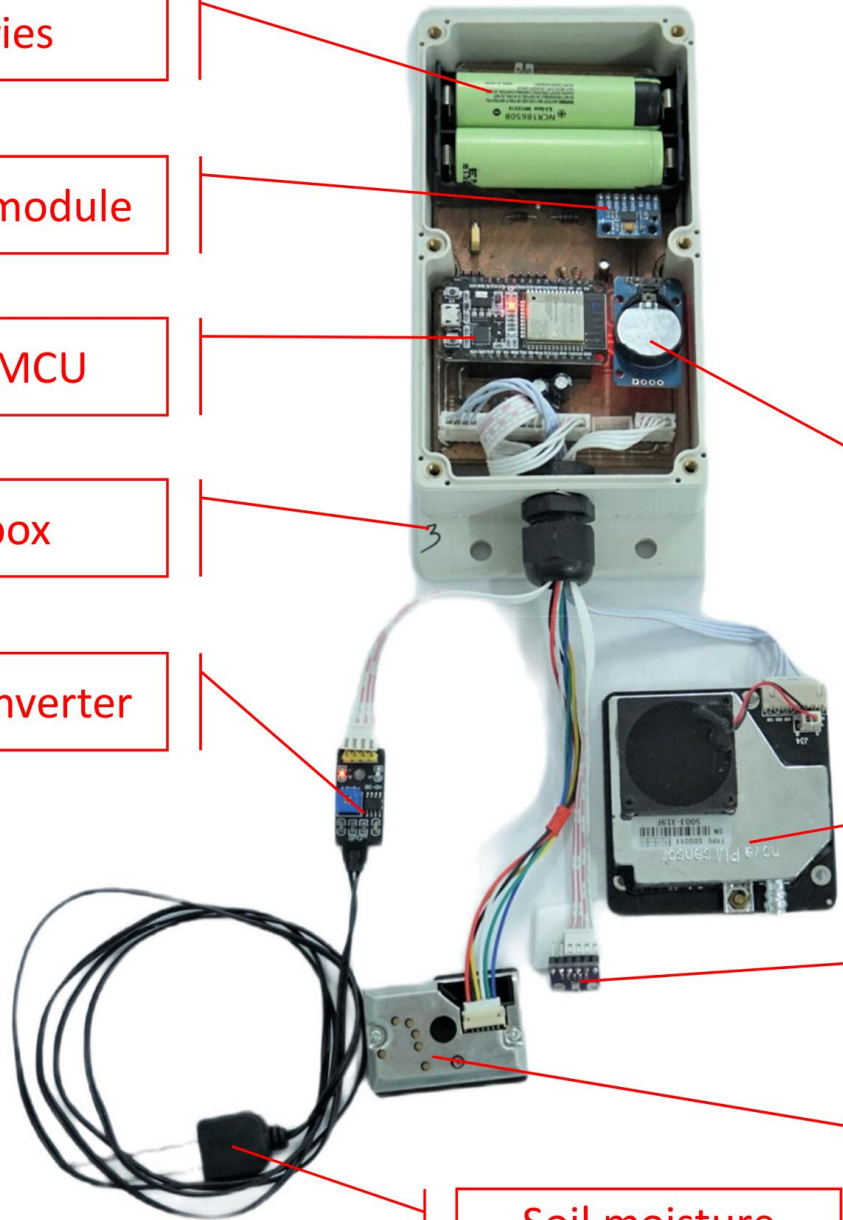
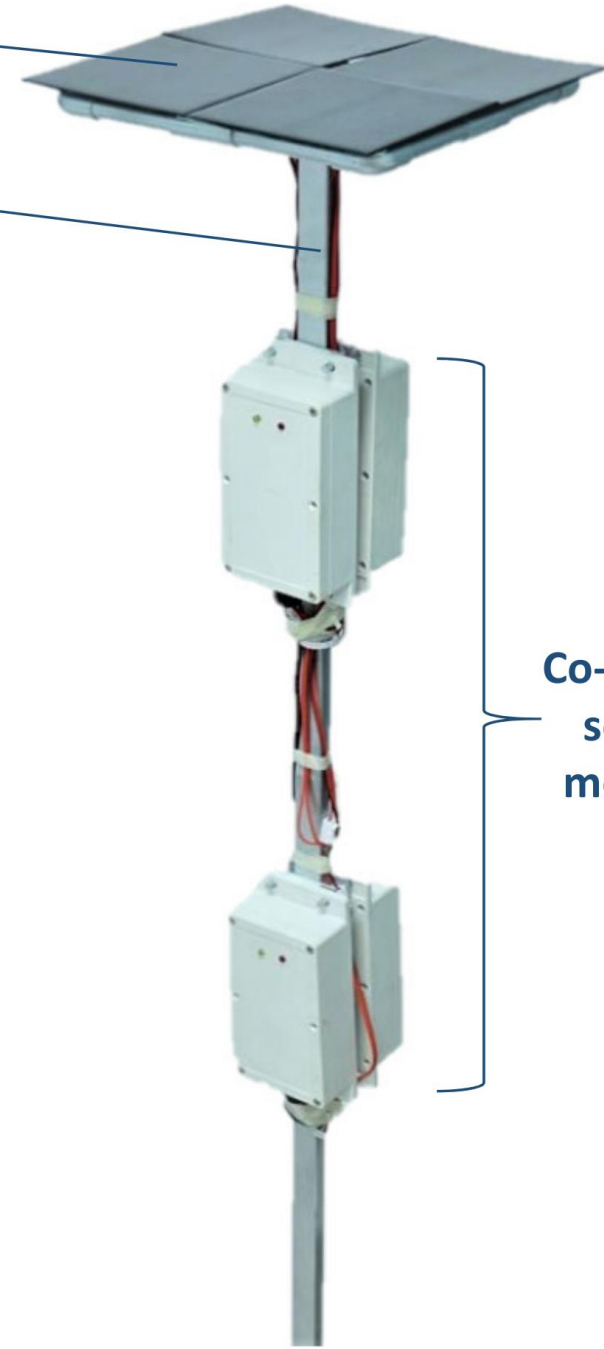
A weather station

Source: Internet

Single Sensor module





Source: Internet

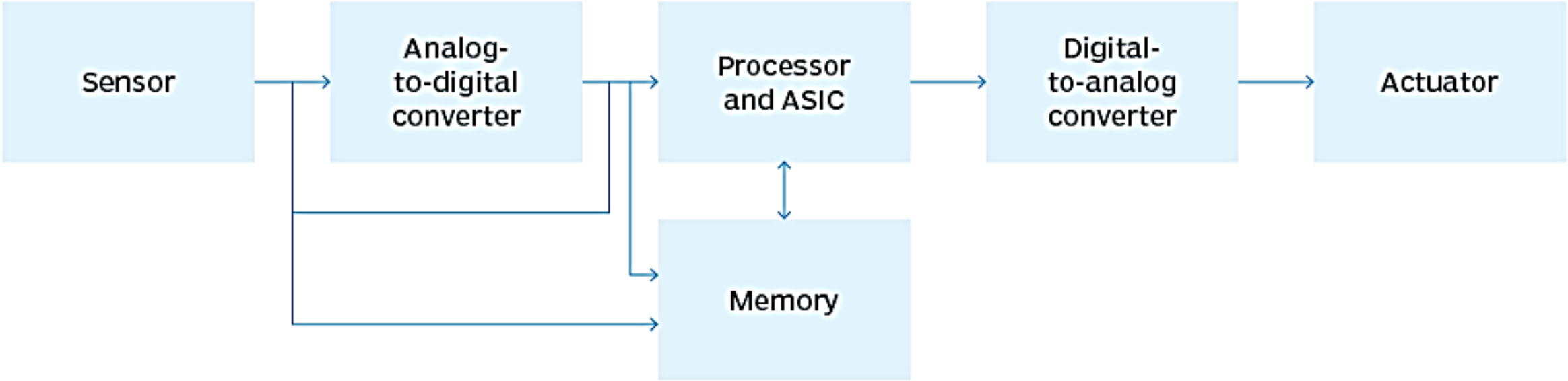
Embedded system

What is an embedded system?

Why ***embedded***?

definition: “(adj) fixed firmly and deeply in a *surrounding mass*; implanted.”

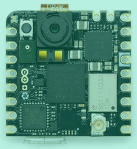
Embedded system structure



Embedded System Components

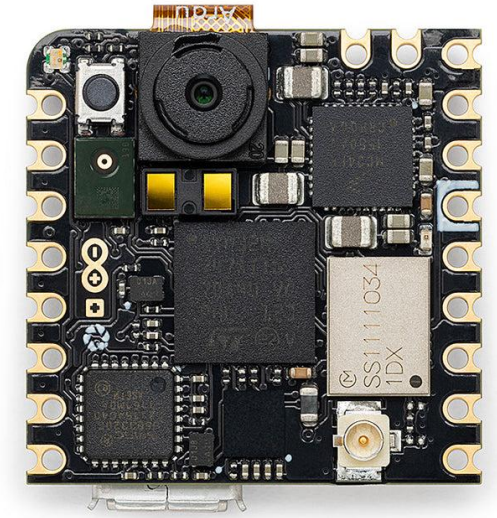
- Microcontroller or Microprocessor
- Memory (RAM and ROM)
- Input Devices (e.g., sensors, switches)
- Output Devices (e.g., LEDs, displays, motors)
- Communication Interfaces (e.g., UART, SPI, I2C, Bluetooth)
- Power Supply
- Timers and Counters
- Analog-to-Digital Converters (ADC) and Digital-to-Analog Converters (DAC)
- Actuators (e.g., motors, solenoids)
- Real-Time Operating System (RTOS) (Optional)
- Embedded Software/Firmware
- Peripherals (e.g., USB ports, GPIO pins)

Example Embedded Platforms

	Board	MCU / ASIC	Clock	Memory	Sensors	Radio
	Himax WE-I Plus EVB	HX6537-A 32-bit EM9D DSP	400 MHz	2MB flash 2MB RAM	Accelerometer, Mic, Camera	None
	Arduino Nano 33 BLE Sense	32-bit nRF52840	64 MHz	1MB flash 256kB RAM	Mic, IMU, Temp, Humidity, Gesture, Pressure, Proximity, Brightness, Color	BLE
	SparkFun Edge 2	32-bit ArtemisV1	48 MHz	1MB flash 384kB RAM	Accelerometer, Mic, Camera	BLE
	Espressif EYE	32-bit ESP32-D0WD	240 MHz	4MB flash 520kB RAM	Mic, Camera	WiFi, BLE
	Arduino Nicla Vision	Dual Arm® Cortex® M7/M4	240 MHz 480 MHz	2MB Flash 1MB RAM	6-Axis IMU 2 MP Color Camera Distance / Time Of Flight sensor Microphone	WiFI, BLE

Arduino Nicla Vision

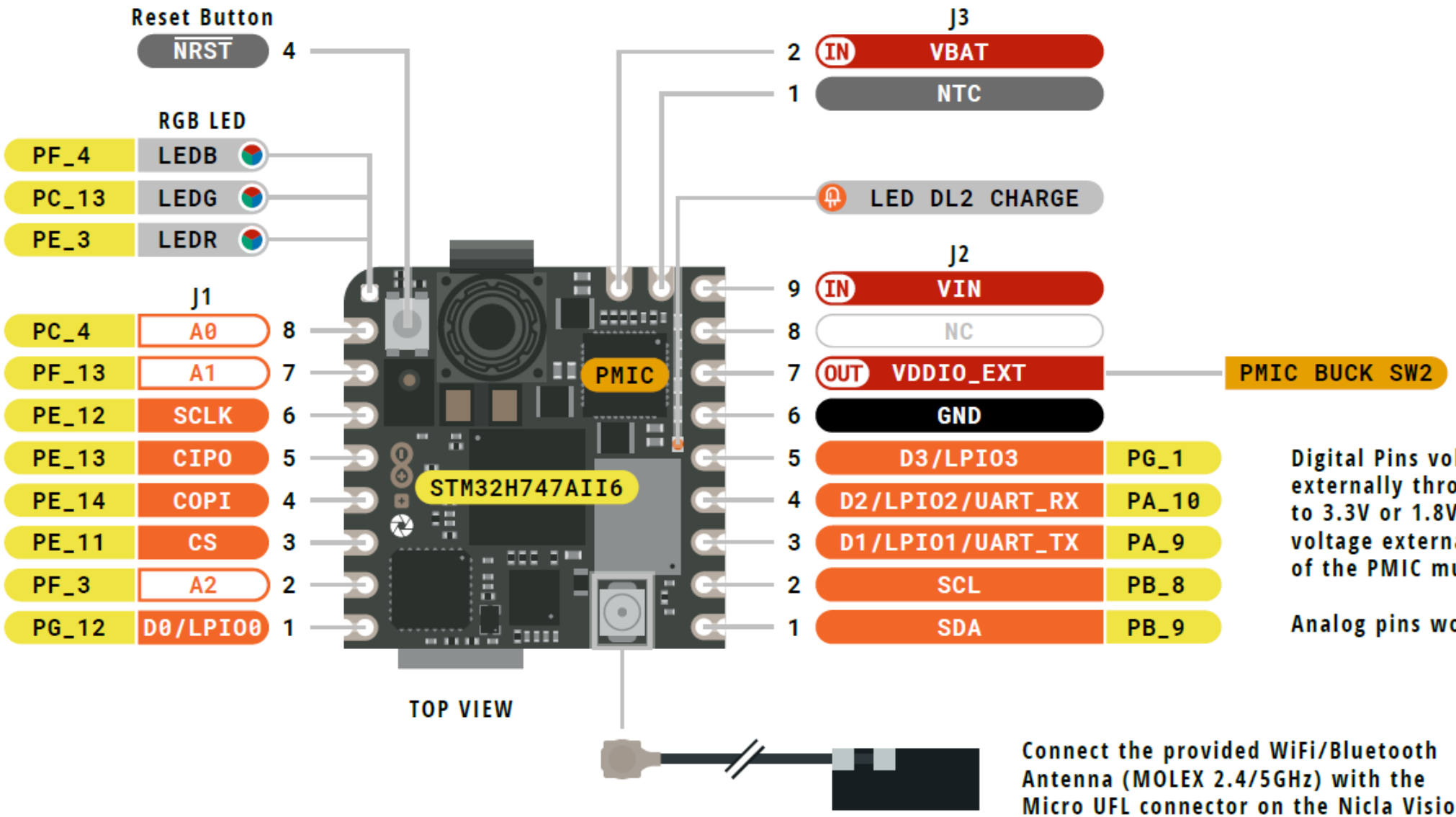
Arduino Nicla Vision



Microcontroller	•STM32H747AII6 Dual Arm® Cortex® M7/M4 IC:1x Arm® Cortex® M7 core up to 480 MHz •1x Arm® Cortex® M4 core up to 240 MHz
Sensors	•2 MP Color Camera •6-Axis IMU (LSM6DSOX) •Distance / Time Of Flight sensor (VL53L1CBV0FY/1) •Microphone (MP34DT05)
Power	•3.7V Li-po battery with Integrated battery charger and fuel gauge (MAX17262REWL)
Memory	2MB Flash / 1MB RAM 16MB QSPI Flash for storage
Connectivity	Wi-Fi / Bluetooth® Low Energy 4.2 (Murata 1DX - LBEE5KL1DX-883)

Source: https://docs.arduino.cc/hardware/nicla-vision/

Arduino Nicla Vision



Digital Pins voltage level can be forced externally through VDDIO_EXT or internally set to 3.3V or 1.8V by software (to force the voltage externally, the buck converter no. 2 of the PMIC must be shut down)

Analog pins work only at 1.8V

Connect the provided WiFi/Bluetooth Antenna (MOLEX 2.4/5GHz) with the Micro UFL connector on the Nicla Vision

Arduino Nicla Vision

- **STM32H747AII6** Microcontroller Dual-core
  - 32-bit Arm® Cortex®-M7 core with double-precision FPU and L1 cache up to 480 MHz
  - 32-bit Arm® 32-bit Cortex®-M4 core with FPU up to 240 MHz
  - Full set of DSP instructions
  - Memory Protection Unit (MPU)

ARM

Cortex®-M processors

MCU + DSP

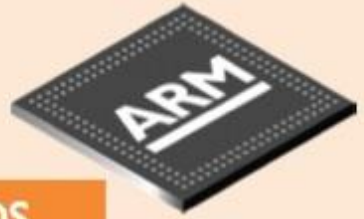
RTOS

Smallest footprint / lowest power

Cortex®-R processors

Highest performance / real-time

Cortex®-A processors

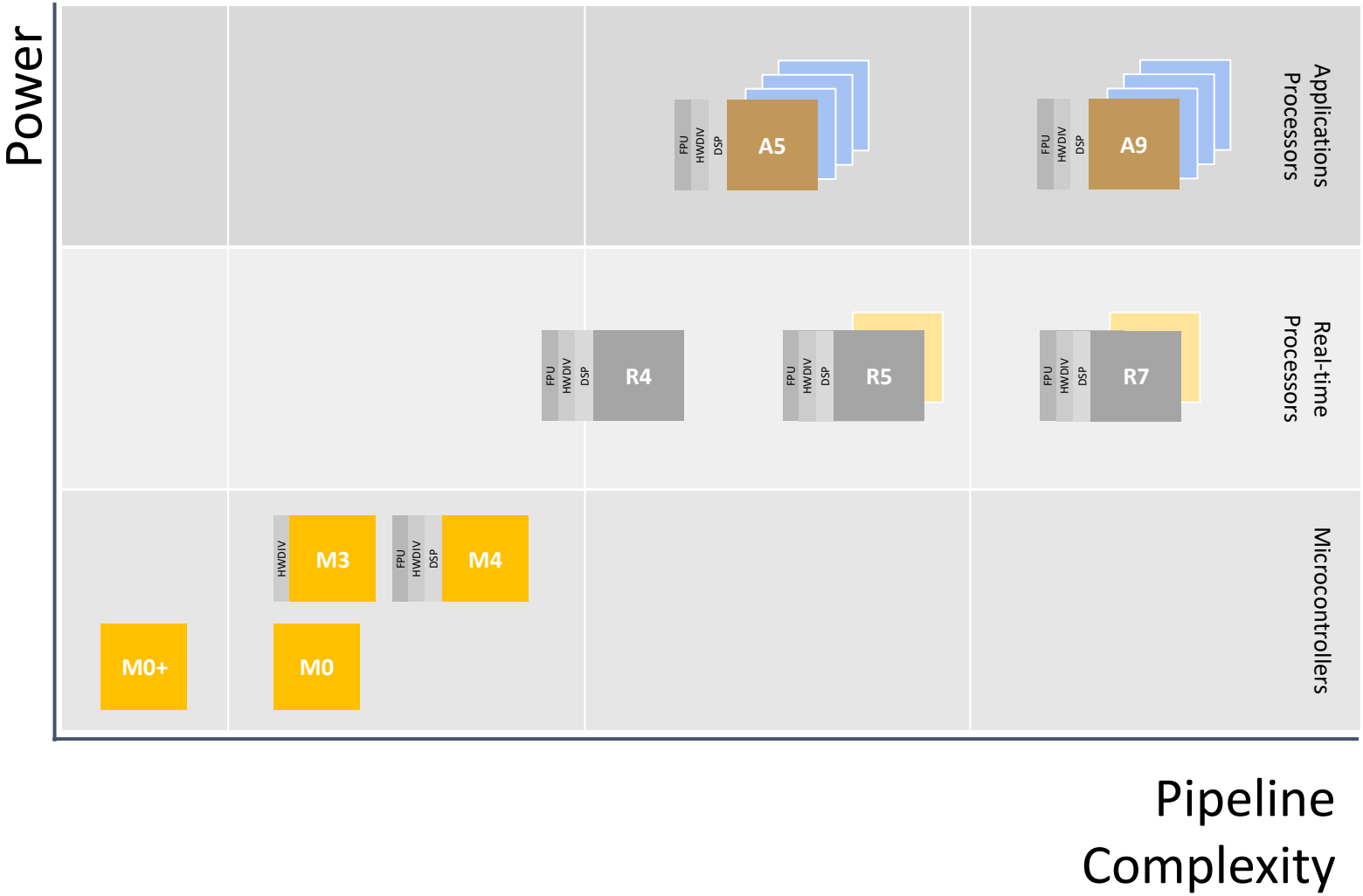


Rich OS

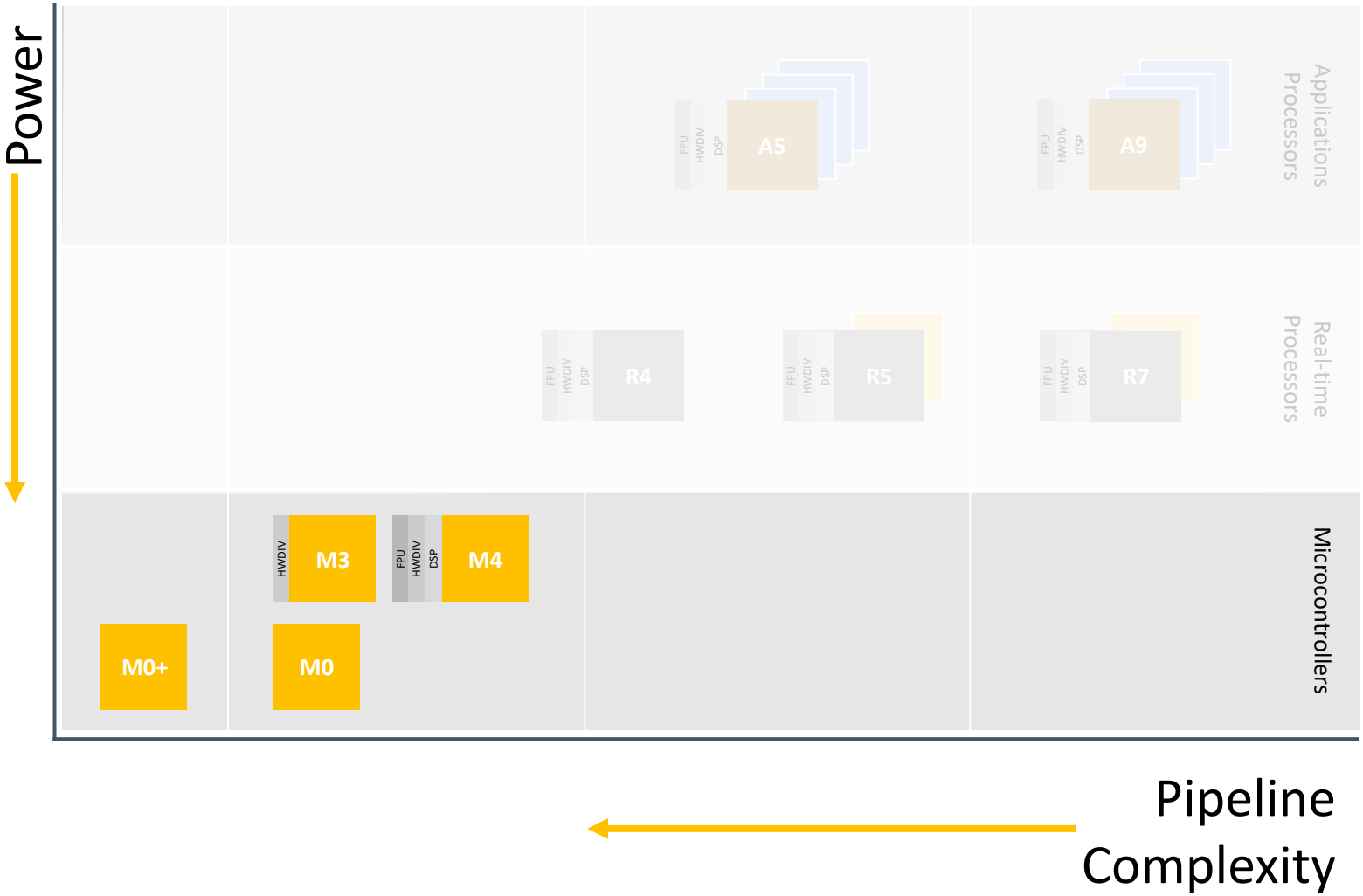
Highest performance

Source: Internet

ARM Cortex Processor Profiles



ARM Cortex Profiles



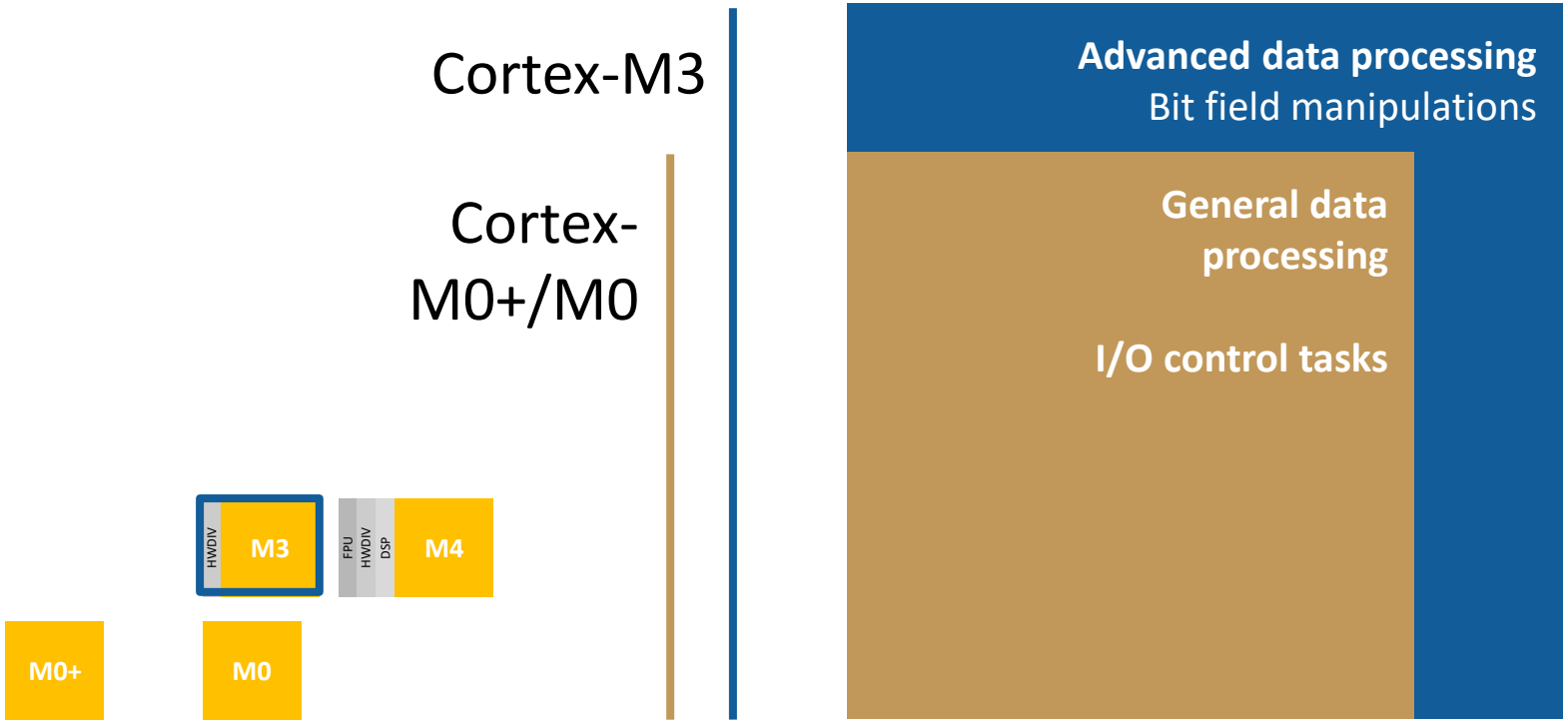
ARM Cortex-M ISA

Cortex-M0+/M0

General data processing

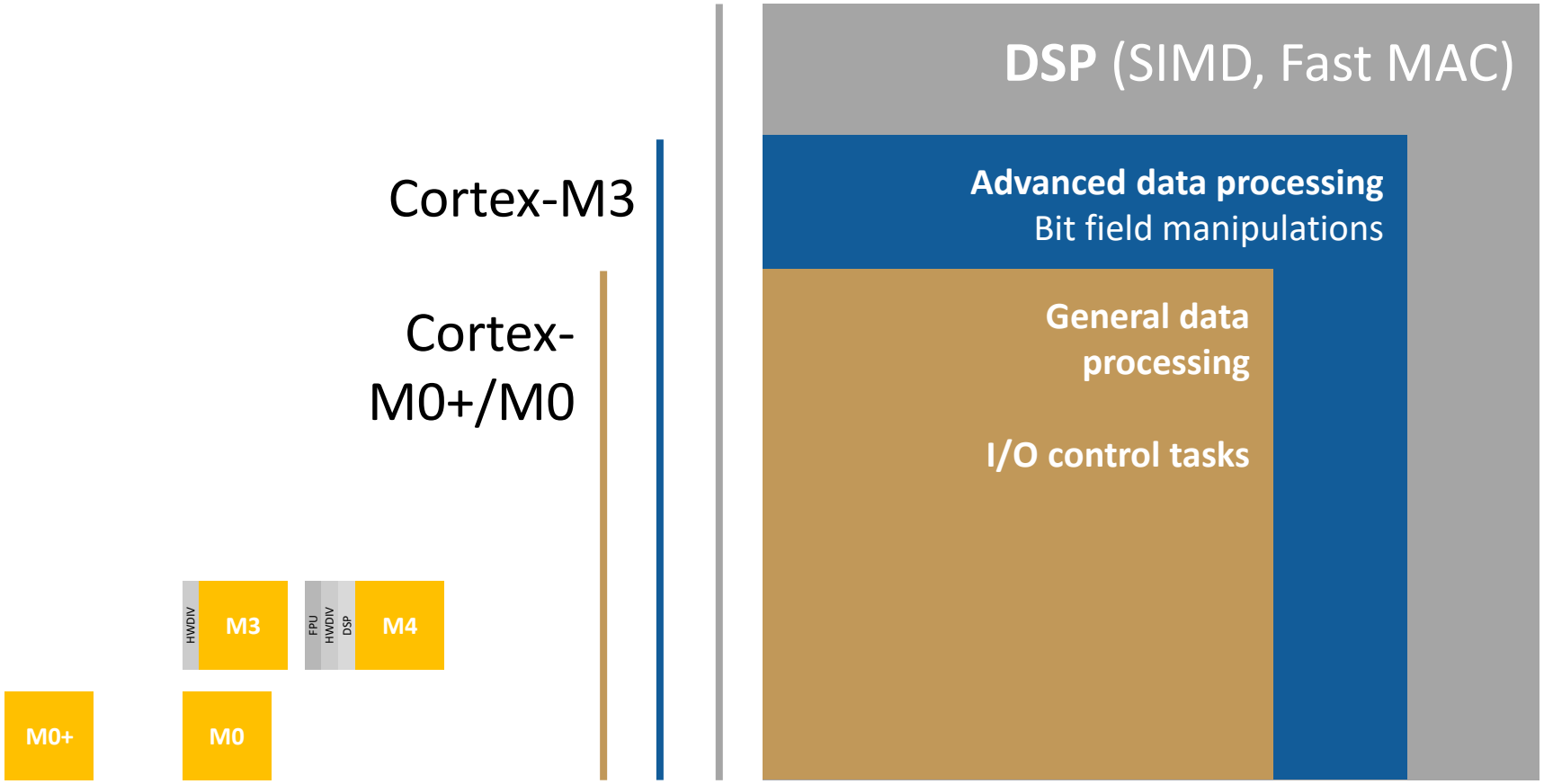
I/O control tasks

ARM Cortex-M ISA



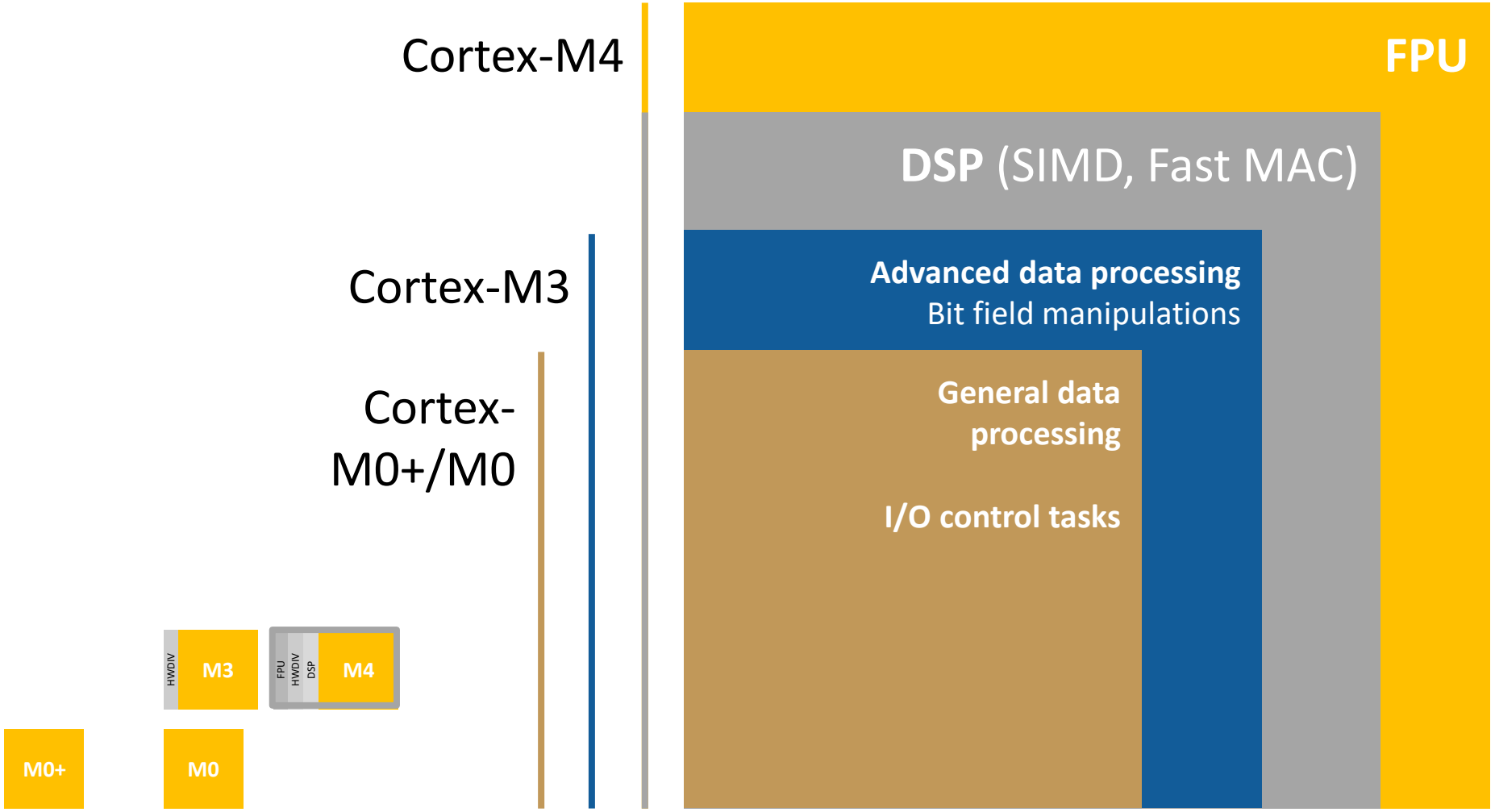
ARM Cortex-M ISA

DSP: Digital signal processing
MAC: multiply and accumulate
SIMD: single instruction multiple data



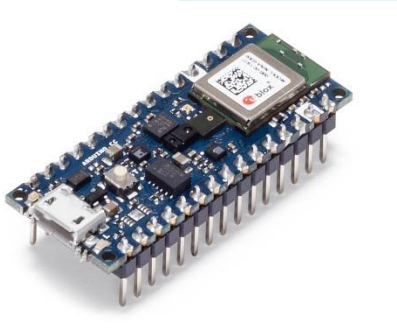
ARM Cortex-M ISA

DSP: Digital signal processing
MAC: multiply and accumulate
SIMD: single instruction multiple data
FPU: floating-point unit



Embedded Software

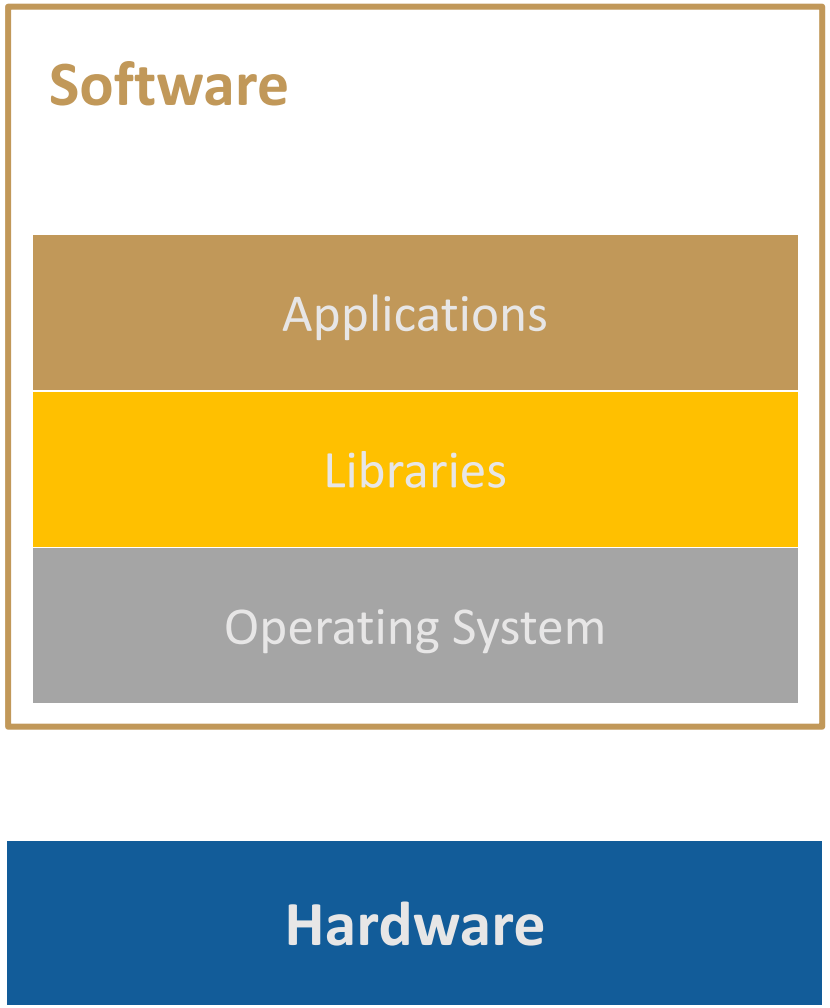
Hardware

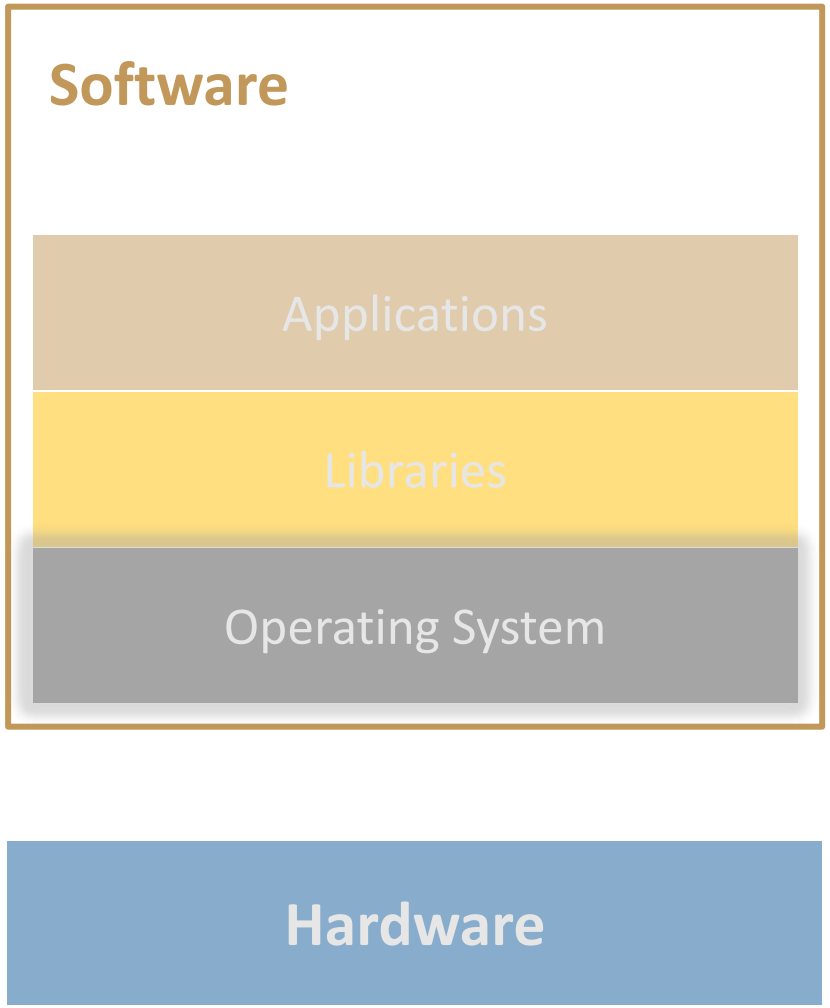


Software

Software

Hardware


Software
Applications
Libraries
Operating System
Hardware

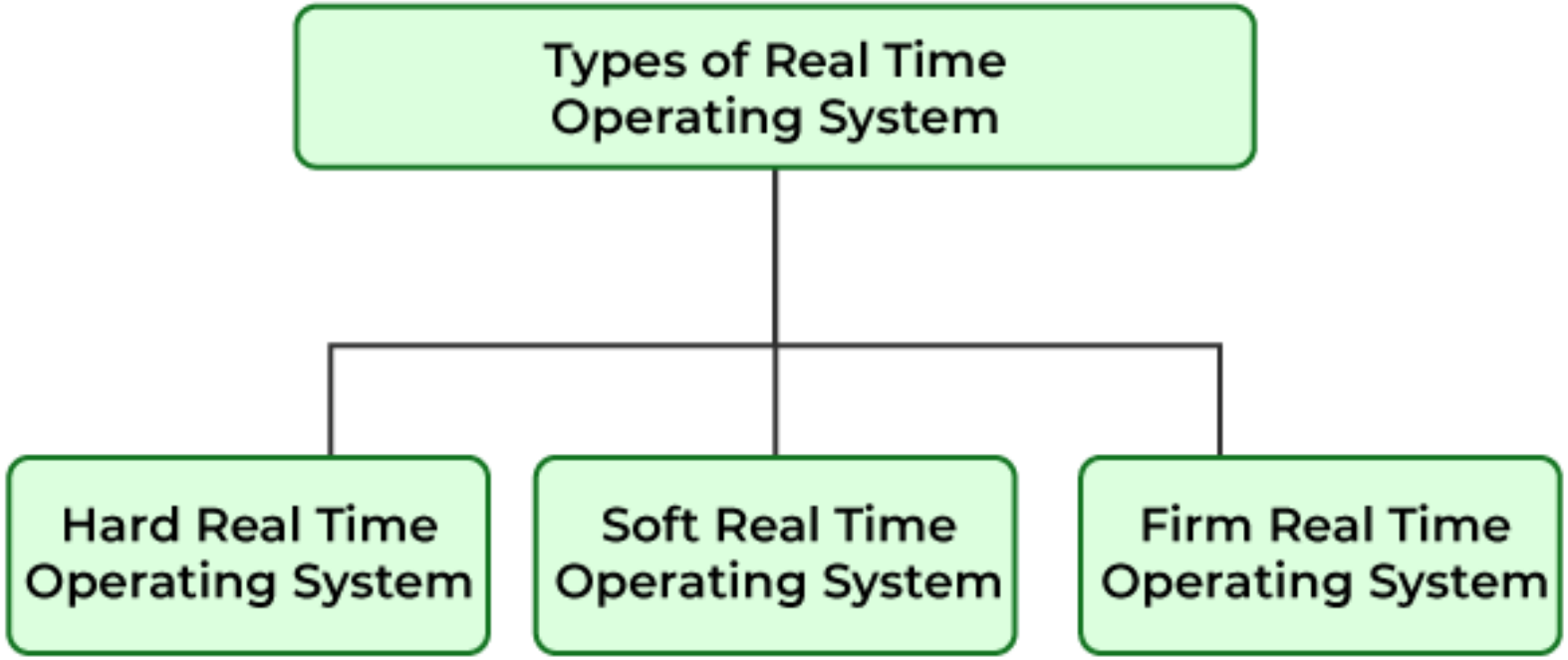

Software
Applications
Libraries
Operating System
Hardware

Widely Used Operating Systems

Mobile OS

Embedded Sys.

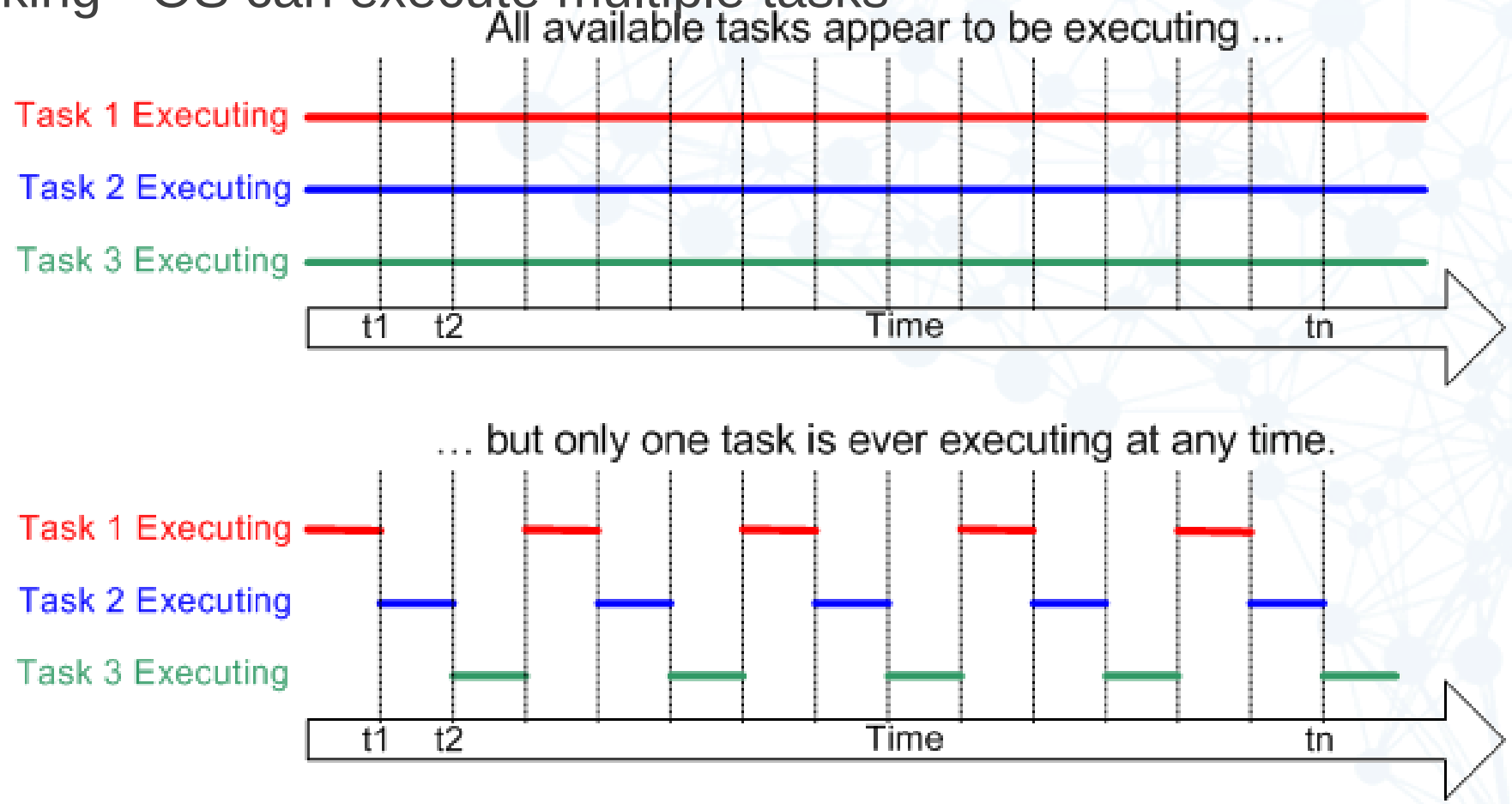
RTOS (Real Time Operating System)



Source: Internet

Multitasking

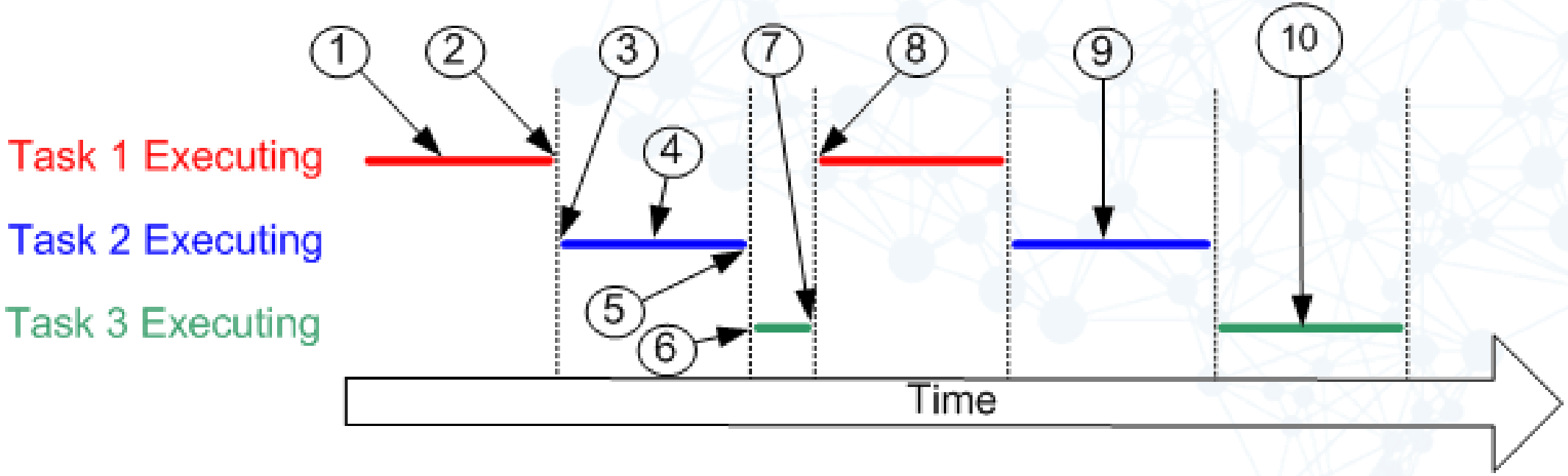
- Task (or thread) - each executing program
- Multitasking - OS can execute multiple tasks



Source: Internet

Scheduling

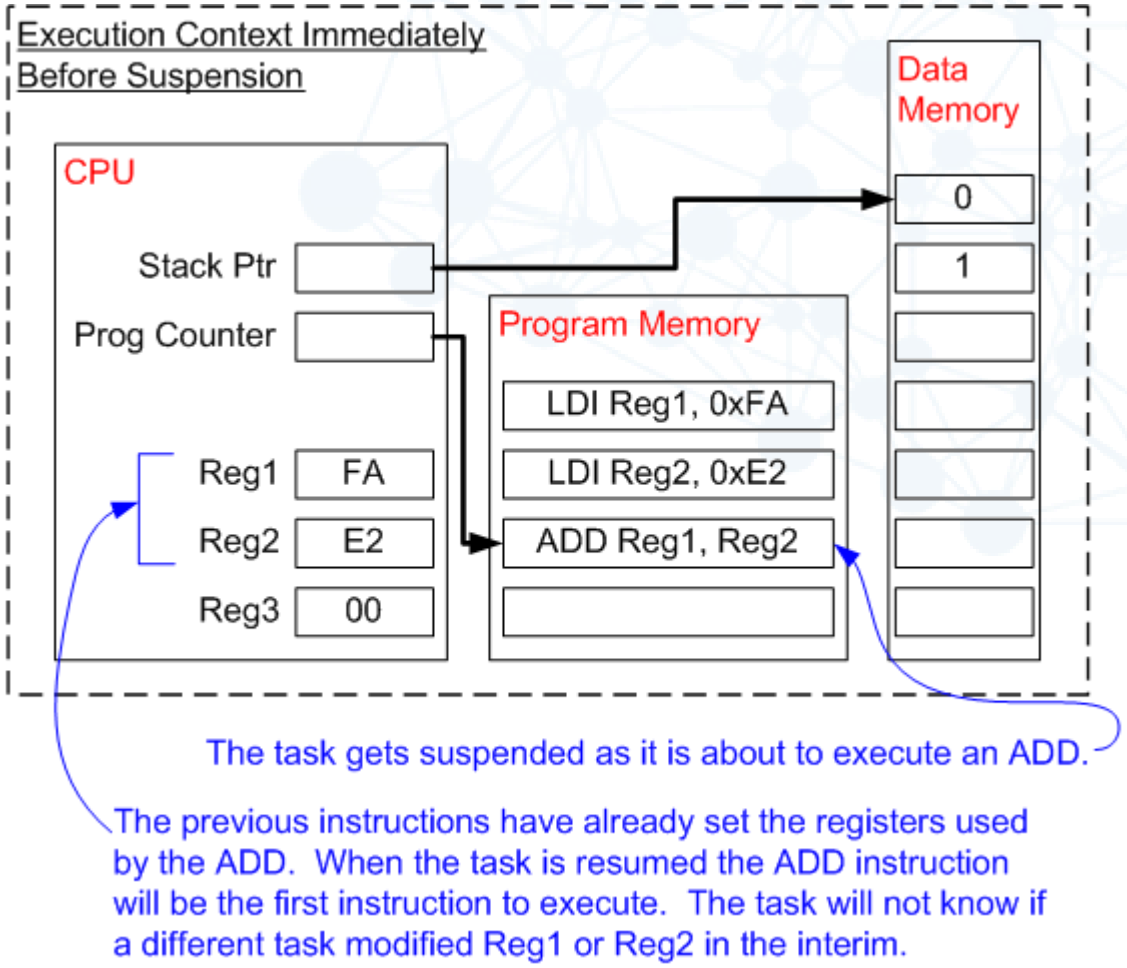
- The scheduler: decides which task should be executing at any particular time



Source: Internet

Context Switching

- Task execution context - microcontroller registers and accesses RAM and ROM

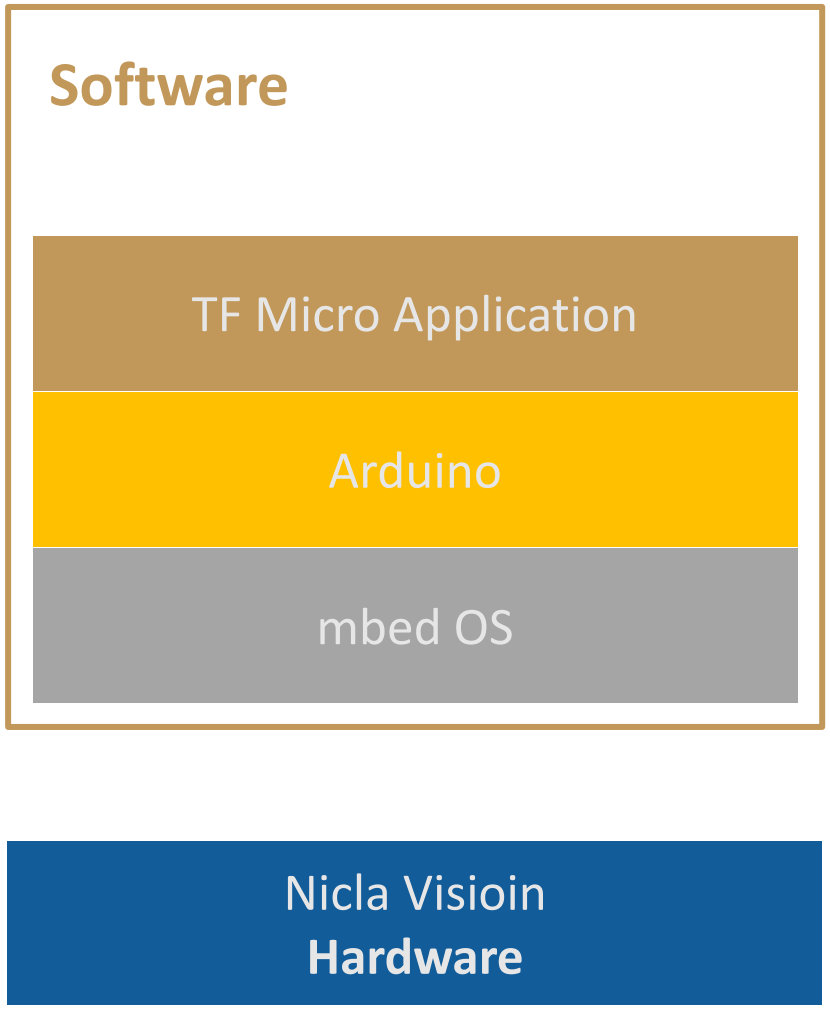


Source: Internet

Widely Used Operating Systems

Mobile OS

Embedded Sys.


Software
TF Micro Application
Arduino
mbed OS
Nicla Visioin
Hardware

Software

TF Micro Application

Arduino

mbed OS

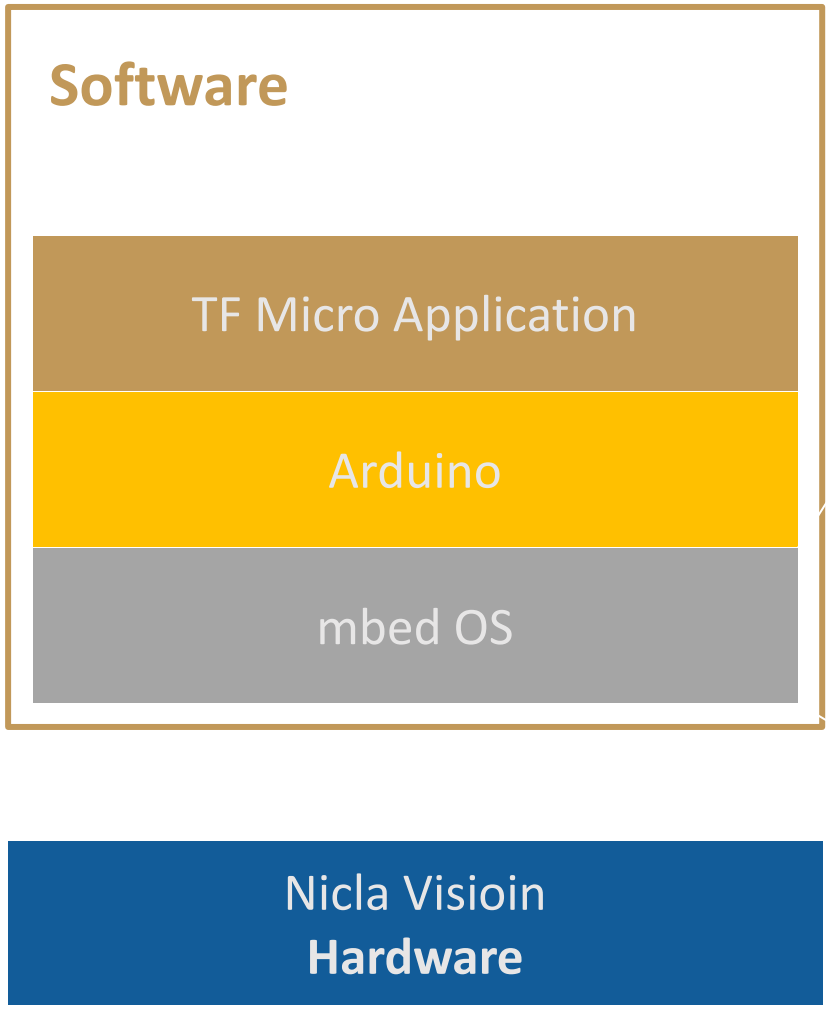
Nicla Visioin
Hardware

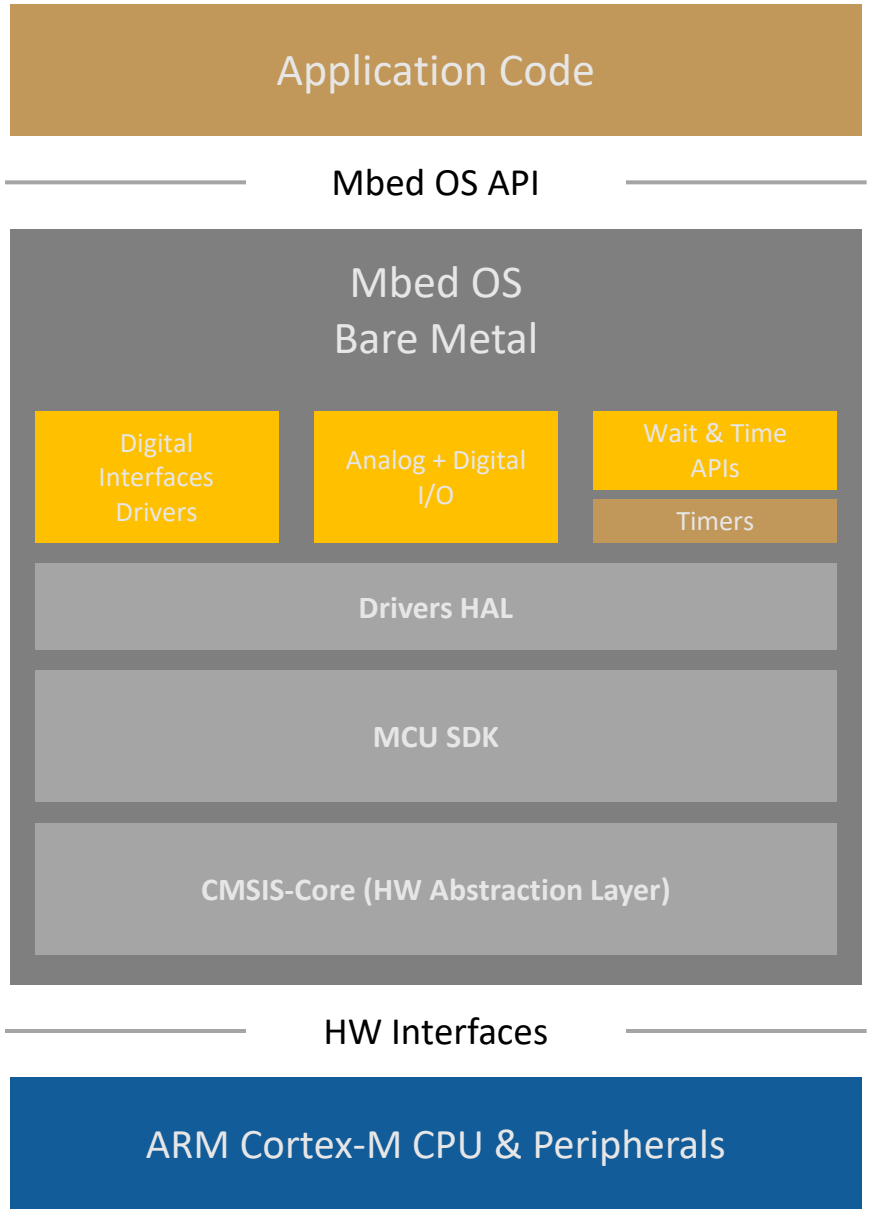
micro_speech | Arduino 1.8.13

micro_speech | arduino_audio_provider.cpp | arduino_command_responder.cpp | arduino_main.cpp

```
#include <TensorFlowLite.h>

#include "main_functions.h"

#include "audio_provider.h"
#include "command_responder.h"
#include "feature_provider.h"
#include "micro_features_micro_model_settings.h"
#include "micro_features_model.h"
#include "recognize_commands.h"
#include "tensorflow/lite/micro/kernels/micro_ops.h"
#include "tensorflow/lite/micro/micro_error_reporter.h"
#include "tensorflow/lite/micro/micro_interpreter.h"
#include "tensorflow/lite/micro/micro_mutable_op_resolver.h"
#include "tensorflow/lite/schema/schema_generated.h"
#include "tensorflow/lite/version.h"

// Globals, used for compatibility with Arduino-style sketches.
namespace {
tflite::ErrorReporter* error_reporter = nullptr;
const tflite::Model* model = nullptr;
tflite::MicroInterpreter* interpreter = nullptr;
TfLiteTensor* model_input = nullptr;
FeatureProvider* feature_provider = nullptr;
RecognizeCommands* recognizer = nullptr;
int32_t previous_time = 0;

// Create an area of memory to use for input, output, and intermediate arrays.
// The size of this will depend on the model you're using, and may need to be
// determined by experimentation.
constexpr int kTensorArenaSize = 10 * 1024;
uint8_t tensor_arena[kTensorArenaSize];
```

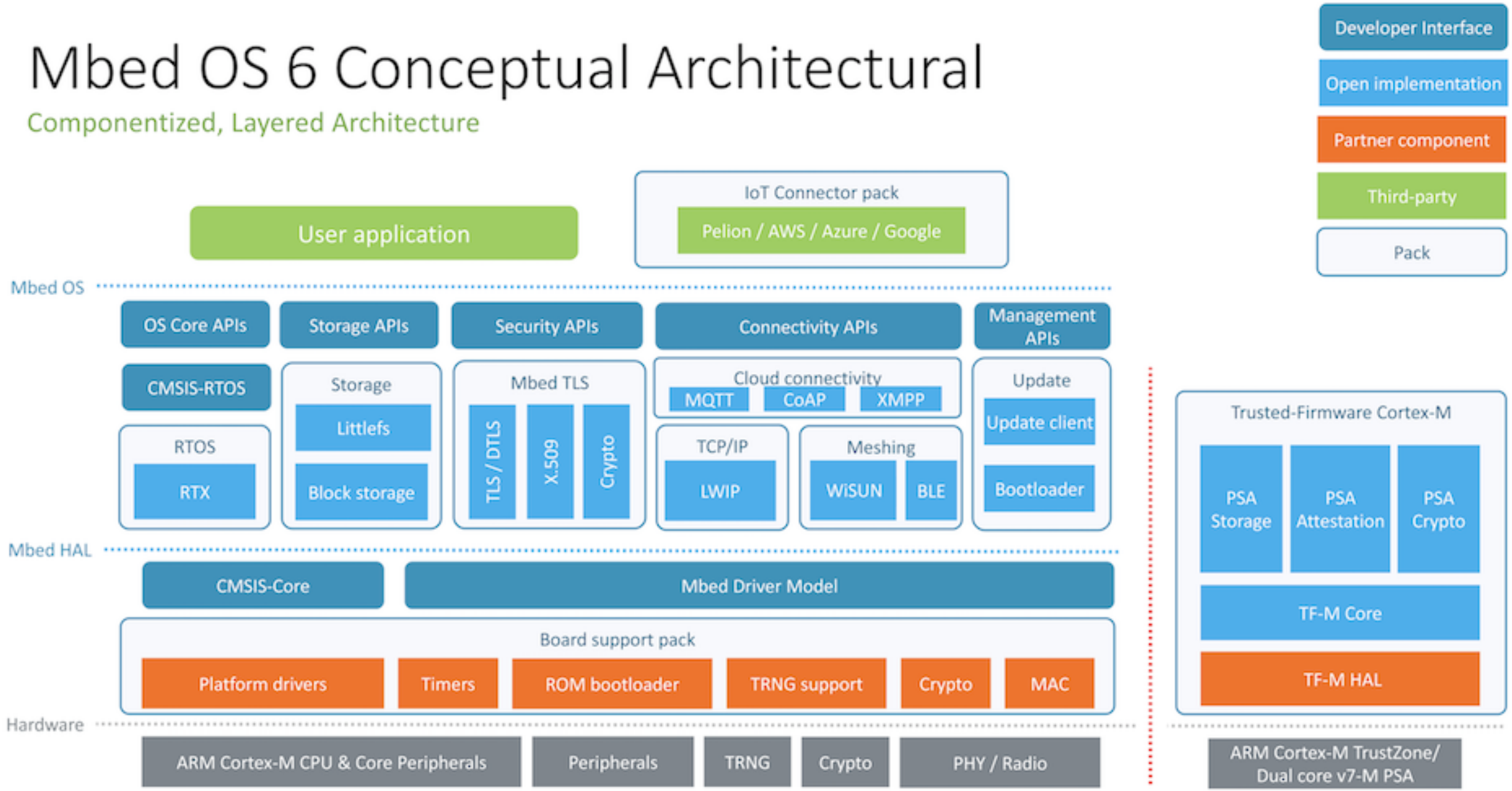
5 Arduino Nano 33 BLE on /dev/cu.usbmodem411101


Software
TF Micro Application
Arduino
mbed OS
Nicla Visioin
Hardware

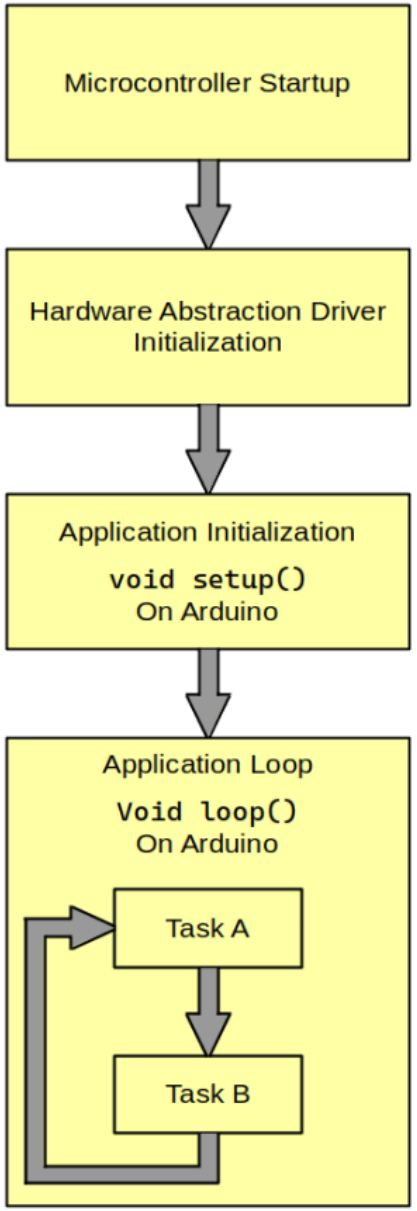

Application Code
Mbed OS API
Mbed OS
Bare Metal
Digital Interfaces Drivers
Analog + Digital I/O
Wait & Time APIs
Timers
Drivers HAL
MCU SDK
CMSIS-Core (HW Abstraction Layer)
HW Interfaces
ARM Cortex-M CPU & Peripherals

Mbed OS 6 Conceptual Architectural

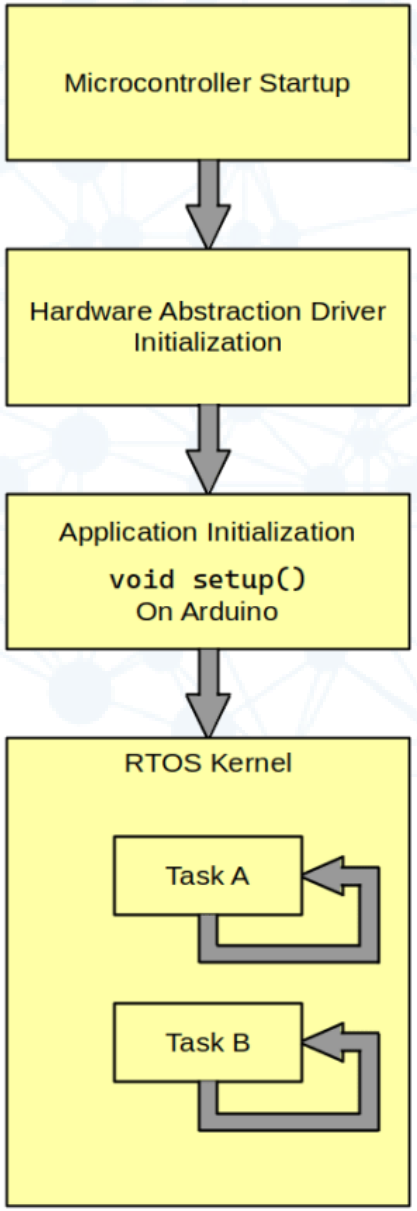
Componentized, Layered Architecture



Arduino Program structure



Mbed OS Program structure



Source: https://dumblebots.com/2020/04/02/using-mbed-with-arduino-arm-boards/

Bare metal profile

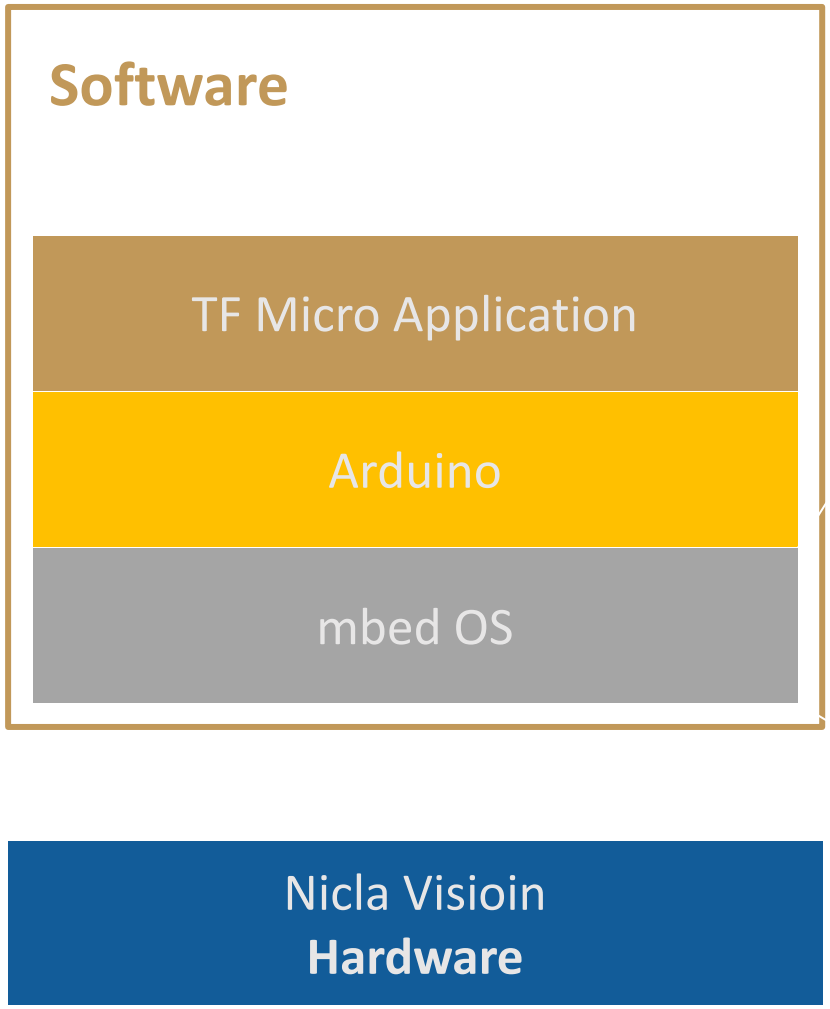
- It is a profile of Mbed OS for ultra constrained hardware
  - compact and without an RTOS

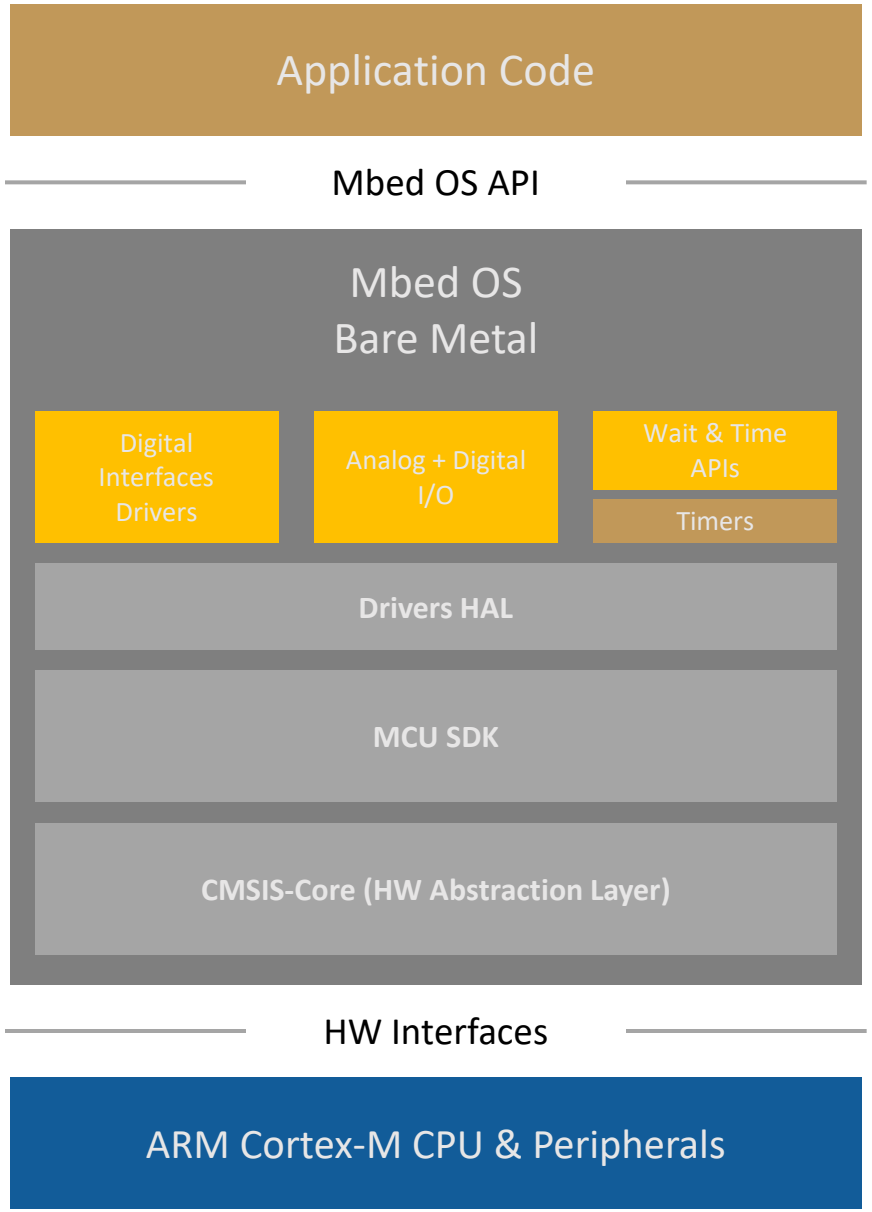
https://os.mbed.com/docs/mbed-os/v6.7/apis/index.html

Features		Support details
Core	Drivers	✓
	Events	✓ (can be manually enabled)
	HAL	✓ (can be manually enabled)
	Platform	✓
	RTOS APIs	Enabled by default: Semaphore, Mutex, EventFlags, ThisThread. All other APIs not supported.
	Storage	✓ (can be manually enabled)
Connectivity	802.15.4_RF	X
	Wifi	X
	Cellular	X
	LWIP stack	X
	Nanostack	X
	Network Socket	X
	BLE	✓ (can be manually enabled) Except the BlueNRG-MS component
	NFC	✓ (can be manually enabled)
Security	PSA	X
	Mbed Crypto	✓ (can be manually enabled)
	Devicekey	✓ (can be manually enabled)
	Mbed TLS	✓ (can be manually enabled)

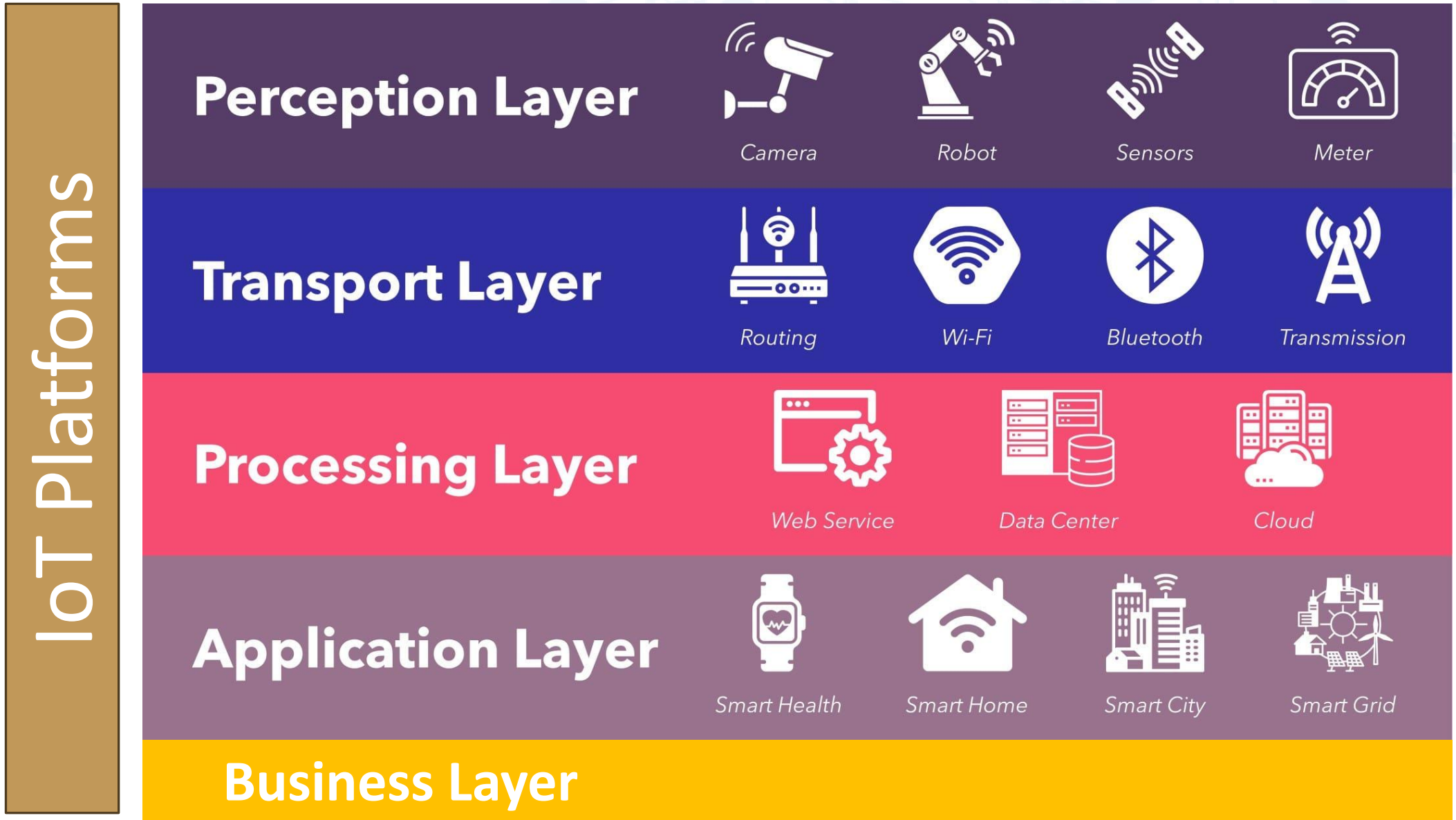
Arduino core

- It is central to a particular microcontroller’s compatibility with the Arduino framework
- arduino.h
  - Declarations for built-in functions. pinMode()
- main.cpp
  - Program structure, setup() and loop()


Software
TF Micro Application
Arduino
mbed OS
Nicla Visioin
Hardware


Application Code
Mbed OS API
Mbed OS
Bare Metal
Digital Interfaces Drivers
Analog + Digital I/O
Wait & Time APIs
Timers
Drivers HAL
MCU SDK
CMSIS-Core (HW Abstraction Layer)
HW Interfaces
ARM Cortex-M CPU & Peripherals

IoT Architecture and Layers



Source: https://dgtlinfra.com/

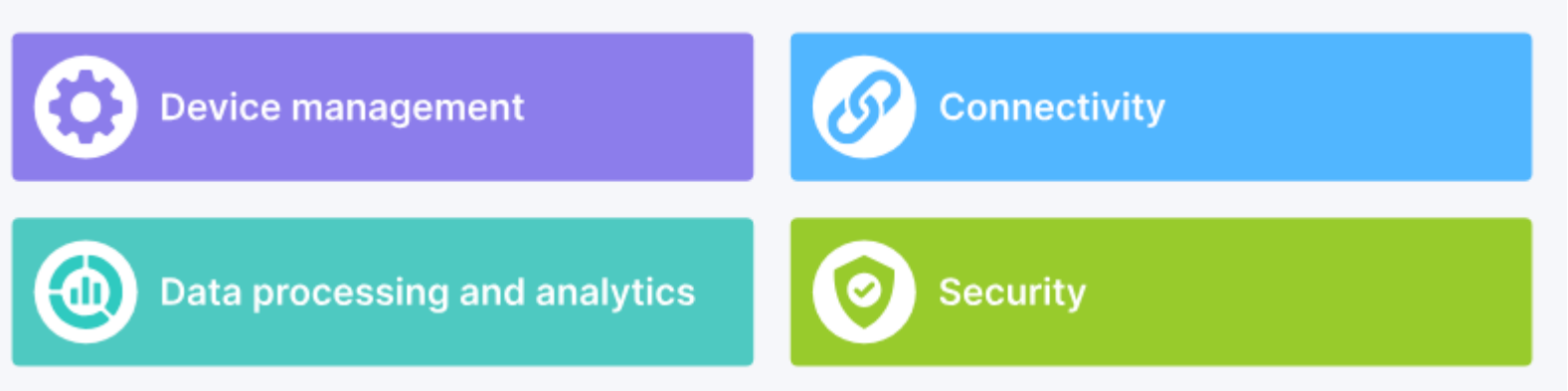
IoT Connectivity

	Short Range	Local Area	Wide Area
Range (typical)	< 10m/30ft	< 100m/300ft	Outdoor (Km/Miles)
Content distribution Focus on high data rates	Bluetooth	WiFi	GSM, LTE, 5G
Sense & control Low energy/long battery life	Bluetooth SMART	ZigBee, GP, ULE Alliance	GPRS, NB-IoT, 5G
Proprietary solutions	ANT+	Z-Wave, enocean	SIGFOX, LoRa, INGENU
Typical applications	**Personal appliance** (wrist band, smart watch, step counter, keybord, mouse, pointer, etc.)	**Idoor networks** (internet, email, phone securitu, energy management, home monitoring, etc.)	**Outdoor networks** (phone, chat, internet, smart city, industry 4.0, agriculture, smart logistics, etc.)

Source: Internet

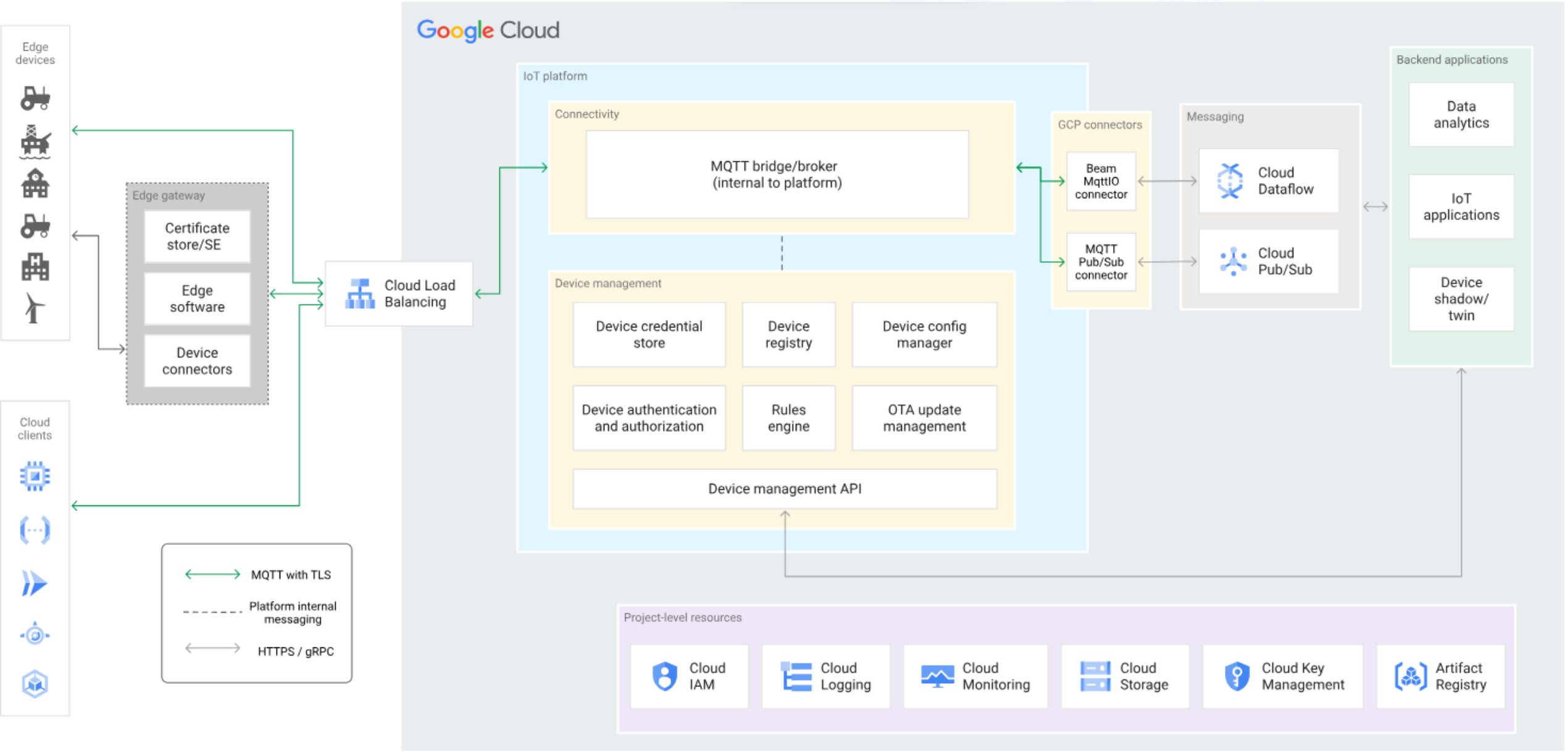
IoT Platforms

- IoT platforms are software solutions that enable the connection, management, and integration of IoT devices, data, and applications to create a unified IoT ecosystem.
- Device Management
- Connectivity Management
- Data Management
- Security and Authentication
- Analytics and Insights



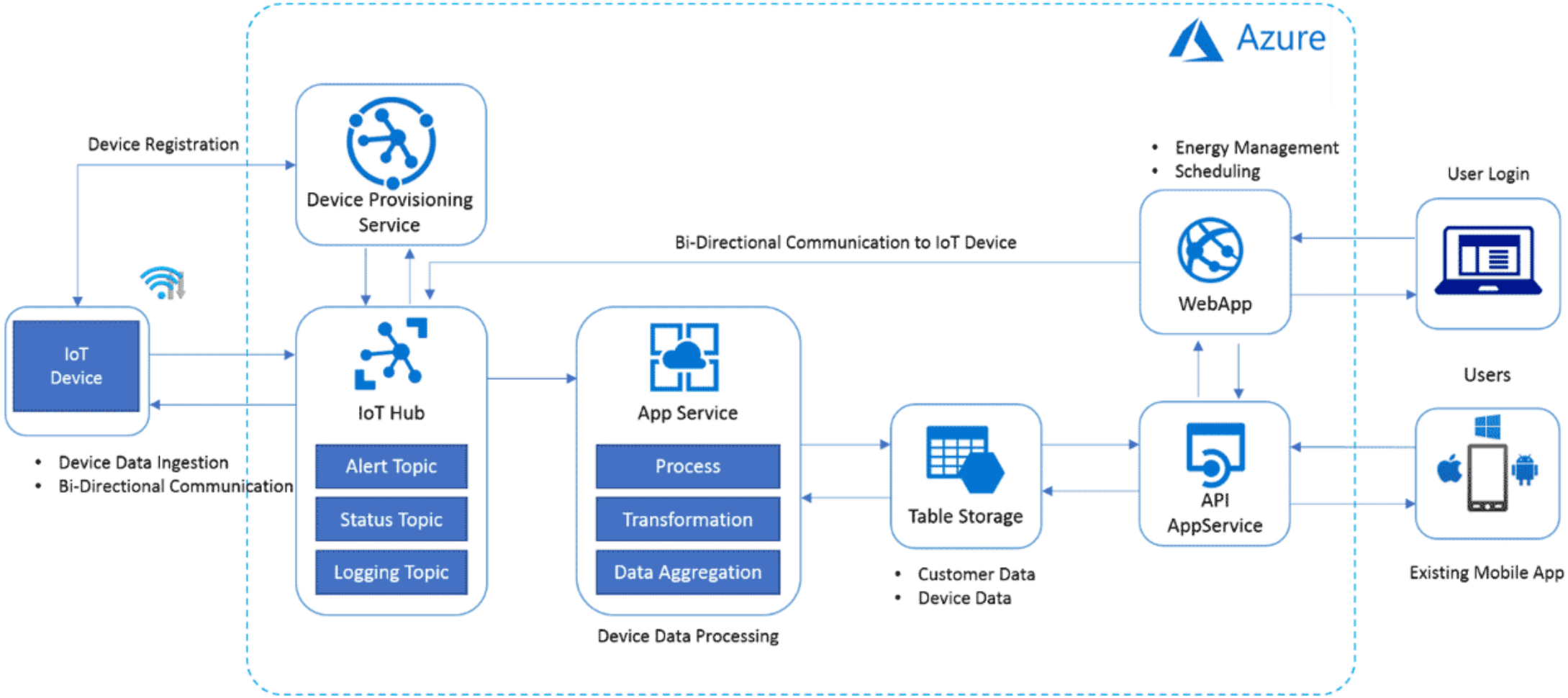
IoT Platforms

- **AWS IoT Core** (Amazon Web Services)
- **Microsoft Azure IoT Hub**
- **Google Cloud IoT Core**
- **IBM Watson IoT Platform**
- **Oracle IoT Cloud**
- **ThingWorx IoT Platform** (PTC)
- **Bosch IoT Suite**
- **Salesforce IoT Cloud**
- **GE Predix**
- **HPE Universal IoT Platform**
- **Samsung Artik Cloud**
- **Cisco IoT Cloud Connect**
- **Alibaba Cloud IoT Platform**
- **Adafruit IO**
- **Particle Cloud**



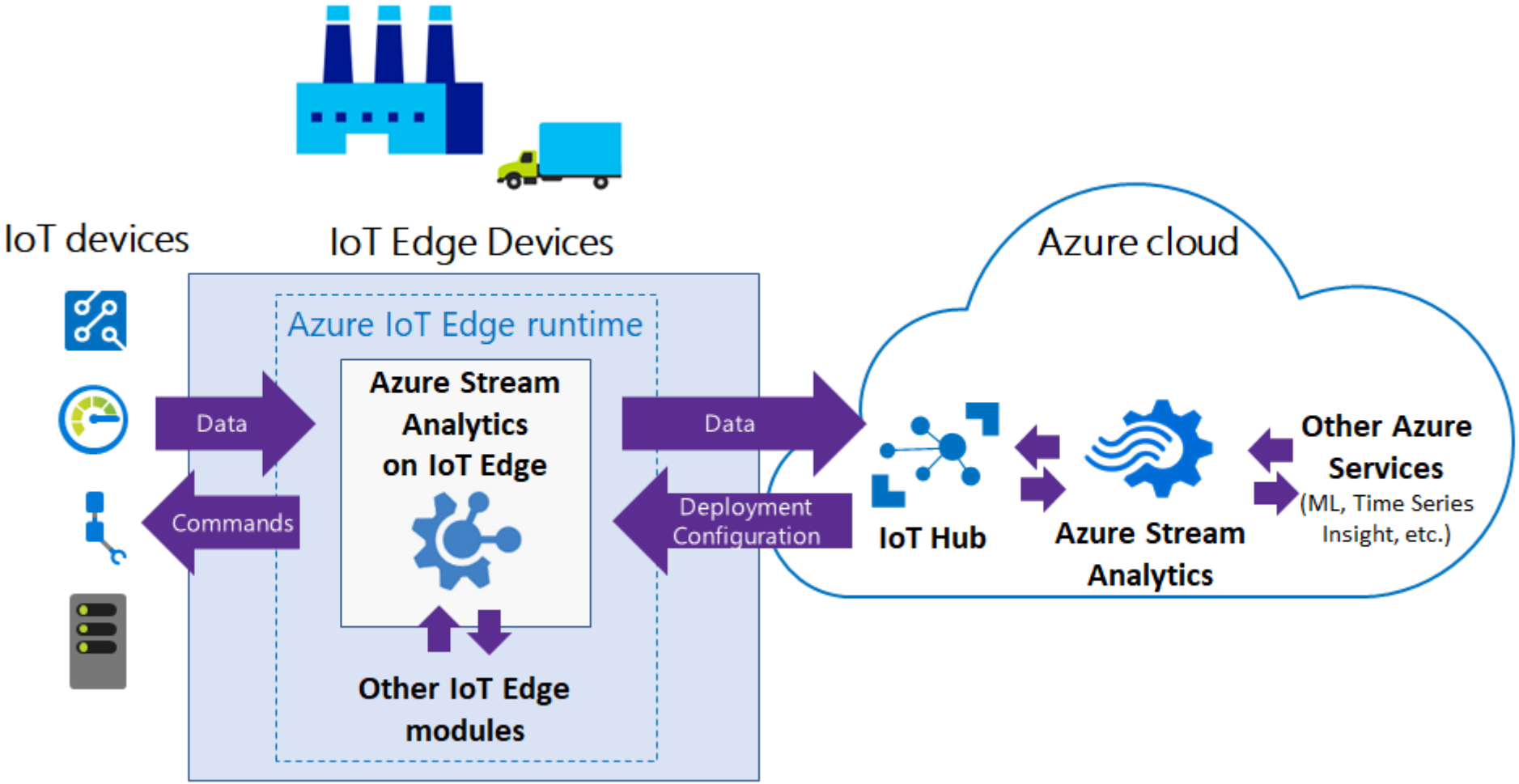
Source: Google

Microsoft Azure IoT



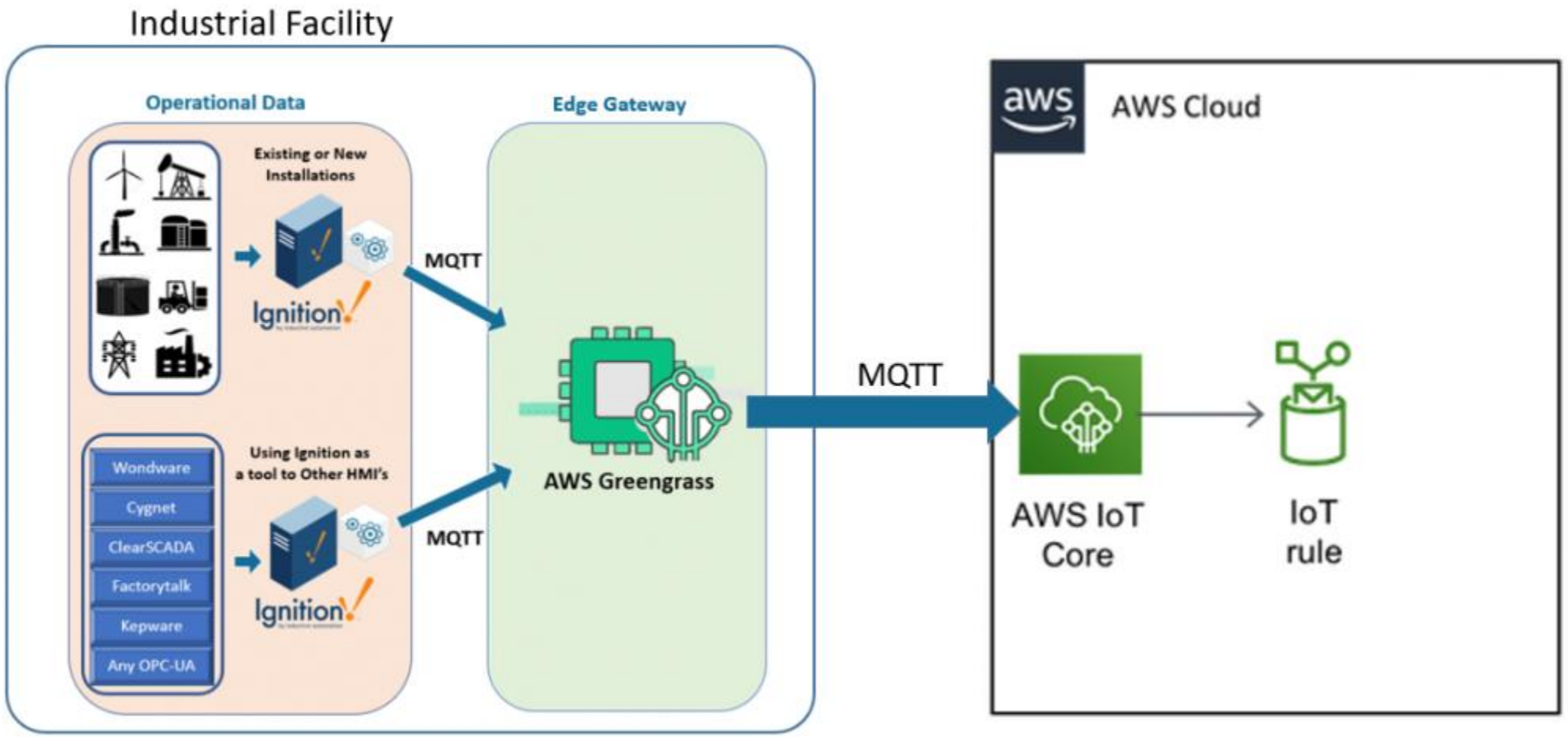
Source: Internet

Microsoft Azure IoT



Source: Internet

AWS IoT


Industrial Facility
Operational Data
Existing or New Installations
Ignition
Using Ignition as a tool to Other HMI's
Wondware
Cygnet
ClearSCADA
Factorytalk
Kepware
Any OPC-UA
Ignition
MQTT
MQTT
Edge Gateway
AWS Greengrass
MQTT
aws
AWS Cloud
AWS IoT Core
IoT rule

IoT Analytics

- IoT Analytics involves processing and analyzing data generated by IoT devices to extract actionable insights and support decision-making.
- **Importance**:
  - Optimizes operations and improves efficiency.
  - Enhances decision-making with data-driven insights.
  - Enables predictive maintenance and reduces downtime.
  - Drives innovation in IoT applications.

IoT Analytics

Collect

Collect device data in a variety of formats and frequencies

Process

Enrich messages with external sources

Analyze

Run queries or use pre-built models to make predictions

Act

Apply findings to improve performance, uptime, efficiency, and other outcomes

Source: Internet

IoT Analytics: Challenges

Data Overload
- Massive volumes of data generated by IoT devices.
- Difficulty in filtering, storing, and processing data efficiently.

Data Quality
- Inconsistent, incomplete, or noisy data from various IoT sensors.
- Ensuring accurate data collection for reliable insights.

Data Integration
- Integrating data from diverse IoT devices, platforms, and systems.
- Handling various data formats, protocols, and standards.

Real-Time Processing
- Need for real-time data processing and analysis.
- Latency issues in processing vast amounts of data quickly.

IoT Analytics: Challenges

Scalability
- Difficulty in scaling IoT systems to handle increasing data and devices.
- Ensuring systems can grow without compromising performance.

Security and Privacy
- Protecting sensitive data transmitted by IoT devices.
- Addressing vulnerabilities in IoT networks and devices.

Edge Computing Challenges
- Offloading data processing to edge devices to reduce latency.
- Managing limited computational resources and power on edge devices.

Analytics Complexity
- Complex algorithms needed to extract meaningful insights from raw IoT data.
- Machine learning models require high-quality labeled data for accurate predictions.

Cost
- High infrastructure and maintenance costs for IoT analytics systems.
- Managing cost-effectiveness while scaling IoT deployments.

Types of IoT Analytics

- **Cloud Analytics**: Data is sent to cloud platforms for centralized processing and analysis.
- **On-Premises Analytics**: Data is processed and analyzed within local infrastructure, typically to meet privacy, security, or regulatory requirements.
- **Edge Analytics**: Data is processed and analyzed locally at the edge of the network, close to IoT devices, reducing latency and dependence on centralized systems.

Cloud Analytics

Cloud Analytics involves sending data from IoT devices to cloud platforms for centralized processing and analysis. This approach leverages powerful cloud computing resources to derive insights from large-scale data.

Example: Predictive maintenance in manufacturing using Microsoft Azure IoT.

Pros:

- Scalable for large datasets and complex analytics.
- Accessible globally through internet connectivity.
- Cost-effective for initial deployment with minimal local infrastructure.
- Easy integration with advanced AI/ML tools for deeper insights.

Cons:

- Higher latency due to data transmission to remote cloud servers.
- Dependent on stable and reliable internet connectivity.
- Potential privacy and security concerns during data transfer.
- Ongoing operational costs for cloud services.

On-Premises Analytics

On-Premises Analytics processes and analyzes data within a local infrastructure, ensuring data privacy and compliance with regulations.

Examples:

- Healthcare systems analyzing patient data within hospital servers.
- Financial institutions processing sensitive transaction data locally.

Pros:

- High security and privacy with local data storage.
- Full control over infrastructure and data.
- Meets stringent regulatory requirements.
- Operates independently of internet connectivity.

Cons:

- Expensive initial setup and maintenance costs.
- Limited scalability compared to cloud solutions.
- Requires skilled personnel for operation and management.
- Higher latency for distributed systems.

Edge Analytics

Edge Analytics processes data locally on IoT devices or nearby edge nodes, reducing dependency on centralized systems.

Examples:

- Autonomous vehicles analyzing sensor data for real-time navigation.
- Smart surveillance cameras performing local video analysis for threat detection.

Pros:

- Low latency due to proximity to data sources.
- Reduces bandwidth usage and reliance on cloud.
- Enables real-time decision-making.
- Scalable for distributed IoT networks.

Cons:

- Limited computational power on edge devices.
- Complex to manage and update across multiple nodes.
- Higher initial cost for robust edge devices.
- May lack advanced analytics capabilities available in the cloud.

Source: https://premioinc.com